

Random Walks and the Gambler's Ruin Problem

Exact Results, Independent Derivations, and an Empirical Application to Equity Index Returns

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19 September 2026

Abstract

This report develops the theory of the simple random walk with two absorbing barriers, the problem known since the seventeenth century as gambler's ruin, and then asks what the model is worth when it is pointed at real market data. The probability of reaching a target before ruin and the expected duration of play are each derived twice, once by solving the governing difference equations through the characteristic polynomial and once by applying the optional stopping theorem to a pair of martingales, and both are then recovered a third time by numerical linear algebra through the fundamental matrix of the absorbing chain. The three routes agree to approximately fifteen decimal places, which is the check on the algebra. Attention is paid to the passage from the asymmetric to the symmetric case, which is usually asserted rather than shown: the limit is obtained by L'Hopital's rule and refined by a series expansion that yields the first-order sensitivity of the ruin probability to a small house edge. A Monte Carlo study of one hundred thousand replications per parameter cell confirms the closed forms to within simulation error, and a convergence study exhibits the expected inverse square root envelope. The applied section estimates the probability of an up day on the S&P 500 composite index over 1990 to 2026 and tests the random walk hypothesis with a runs test, a Ljung-Box test and the Lo-MacKinlay variance ratio test. The sign sequence is found to be statistically distinguishable from an independent sequence, though the effect is small in magnitude. Applying the ruin formulae with the estimated win probability gives a modelled ruin probability that the report then shows to be badly optimistic: replacing the fixed one unit stake with a bootstrap of the actual return distribution raises the projected probability of ruin from roughly five per cent to roughly forty per cent. The gap between those two numbers is the central applied finding, and it is a caution about the fixed step size assumption rather than about the mathematics.

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1 Introduction

A gambler enters a casino with a fixed sum, stakes one unit on each of a sequence of independent wagers, and leaves either when the money is gone or when a self-imposed target has been reached. Two questions present themselves immediately. What is the chance that the target is reached rather than the money exhausted, and how long should the gambler expect to be at the table? These are the questions of the gambler’s ruin problem, and their answers have a closed form that is elementary to state and surprisingly sharp in its consequences.

The problem has a documented origin. In 1654 Pascal and Fermat exchanged the letters on the division of stakes that are conventionally taken to mark the beginning of mathematical probability. The ruin problem proper appears a little later, in 1656, when Pascal posed it to Fermat through the intermediary Pierre de Carcavi, and it reached print in 1657 as the fifth and last of the exercises appended to Huygens’ *De Ratiociniis in Ludo Aleae* (Huygens 1657), the first published treatise on probability. Huygens stated the answer without proof. In his formulation two players, A and B, hold twelve counters each and throw three dice repeatedly; A gains a counter whenever eleven is thrown and B gains one whenever fourteen is thrown, all other throws being disregarded, and play continues until one player holds all twenty-four counters. Huygens gave the odds on A as **282,429,536,481** to **244,140,625**. Section 3.6 reproduces that figure from the general formula derived below, which is a pleasant check on both the formula and the seventeenth century arithmetic. Accounts of the early history are given by Edwards (1983), Takács (1969) and Hald (2003).

The modern treatment runs through the theory of Markov chains and martingales, and the canonical reference remains Chapter XIV of Feller (1968). The problem earns its continued attention for three reasons. It is one of the few non-trivial stochastic models with a complete and exact solution, so it serves as a proving ground for technique. It is the discrete skeleton of Brownian motion with drift, and therefore sits directly upstream of the continuous time models used in derivative pricing (Bachelier 1900; Black and Scholes 1973). And it delivers a conclusion about risk that is both quantitative and counter-intuitive, namely that a player of finite means facing an opponent of effectively unlimited means is ruined with certainty whenever the odds are not in the player’s favour, however slightly.

This report has two halves. The first, Sections 2 to 9, is mathematical. It states the model formally, derives the ruin probability and the expected duration from the governing difference equations, re-derives the ruin probability independently using martingales and the optional stopping theorem, recovers both results numerically through the fundamental matrix of the absorbing chain, examines the limiting behaviour as the barrier recedes and as the game approaches fairness, sketches the diffusion limit, and validates everything by simulation. The second half, Section 10, applies the machinery to daily returns on the S&P 500 composite index, tests whether the sign sequence of those returns is consistent with the independence the model assumes, and then quantifies how much the model’s answer changes when its fixed stake assumption is replaced by the empirical distribution of returns. Section 11 sets out the limitations and four directions for extension.

A word on how the numbers in this document were produced. Every quantity quoted in the text is generated by an inline expression evaluated when the document is compiled, so no figure in the prose can drift out of step with the computation that produced it. Stochastic sections are seeded. The analytical, numerical and simulated answers are compared programmatically at several points. Each comparison is recorded as a verification check with an explicit tolerance, and any quantity whose check fails is withheld from the text rather than printed, so a number that has not passed its own verification cannot appear here looking as though it had. Section 12 lists every check, its tolerance and its outcome. The full code is reproduced in Appendix A and the computing environment in Appendix B.

2 Formal setup and Markov chain formulation

2.1 The model

Let $p \in (0, 1)$ denote the probability that the gambler wins a single round and let $q = 1 - p$ denote the probability of losing it. Let N be a fixed integer with $N \geq 2$, the target capital, and let the gambler’s initial

capital be an integer i with $0 \leq i \leq N$. Let $X_1, X_2, \dots$ be independent and identically distributed random variables on $\{+1, -1\}$ with

$$\mathbb{P}(X_k = +1) = p, \quad \mathbb{P}(X_k = -1) = q = 1 - p,$$

interpreted as the gain in units of capital on round k . The gambler's capital after n rounds is the partial sum

$$S_n = i + \sum_{k=1}^n X_k, \quad S_0 = i, \quad (1)$$

stopped at the first time it reaches either boundary. Formally, define the absorption time

$$\tau = \inf\{n \geq 0 : S_n = 0 \text{ or } S_n = N\}, \quad (2)$$

with the convention $\inf \emptyset = \infty$, and take $S_n = S_\tau$ for all $n \geq \tau$. The process $(S_n)_{n \geq 0}$ so defined is the simple random walk on $\{0, 1, \dots, N\}$ absorbed at both endpoints. The two quantities of interest are

$$P_i = \mathbb{P}(S_\tau = N \mid S_0 = i), \quad D_i = \mathbb{E}[\tau \mid S_0 = i],$$

the probability of reaching the target before ruin, and the expected duration of play. The complementary probability of ruin is $1 - P_i$, and the report writes it that way rather than introducing a second symbol.

2.2 The chain and its transition matrix

Because the increments are independent of the past and the increment distribution does not depend on the current capital, (S_n) is a time homogeneous Markov chain on the finite state space $\mathcal{S} = \{0, 1, \dots, N\}$. Its transition matrix $P = (P_{jk})_{j,k \in \mathcal{S}}$, where $P_{jk} = \mathbb{P}(S_{n+1} = k \mid S_n = j)$, has entries

$$P_{jk} = \begin{cases} p, & 1 \leq j \leq N-1 \text{ and } k = j+1, \\ q, & 1 \leq j \leq N-1 \text{ and } k = j-1, \\ 1, & j = k = 0 \text{ or } j = k = N, \\ 0, & \text{otherwise,} \end{cases} \quad (3)$$

so that written out in full the matrix is tridiagonal apart from the two absorbing rows,

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ q & 0 & p & 0 & \cdots & 0 & 0 \\ 0 & q & 0 & p & \cdots & 0 & 0 \\ \vdots & & \ddots & \ddots & \ddots & & \vdots \\ 0 & 0 & \cdots & q & 0 & p & 0 \\ 0 & 0 & \cdots & 0 & q & 0 & p \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix}. \quad (4)$$

States 0 and N are absorbing, since $P_{00} = P_{NN} = 1$. Every state in $\{1, \dots, N-1\}$ is transient, a fact that follows from Lemma 2.1 below. Table 1 displays the matrix for a small case so that its structure is visible.

Table 1: Transition matrix for $N = 6$ and $p = 0.6$. The first and last rows are absorbing; every interior row places mass p one step up and q one step down. Zeros are shown to make the tridiagonal band explicit.

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
$j = 0$	1.0	0.0	0.0	0.0	0.0	0.0	0.0
$j = 1$	0.4	0.0	0.6	0.0	0.0	0.0	0.0
$j = 2$	0.0	0.4	0.0	0.6	0.0	0.0	0.0
$j = 3$	0.0	0.0	0.4	0.0	0.6	0.0	0.0
$j = 4$	0.0	0.0	0.0	0.4	0.0	0.6	0.0
$j = 5$	0.0	0.0	0.0	0.0	0.4	0.0	0.6
$j = 6$	0.0	0.0	0.0	0.0	0.0	0.0	1.0

The following lemma is the technical fact that makes everything else legitimate. It is stated separately because both the martingale argument of Section 5 and the claim that the interior states are transient depend on it, and because it is the step most often passed over in silence.

Lemma 2.1. *For every $p \in (0, 1)$ and every starting state $i \in \{0, 1, \dots, N\}$, the absorption time τ of eqref:tau is almost surely finite, and moreover $\mathbb{E}[\tau] < \infty$ and τ has a geometrically bounded tail.*

Proof. Set $m = \min(p, q) > 0$. Consider the walk in consecutive blocks of N rounds. Whatever the capital at the start of a block, if all N increments in that block equal $+1$ then the capital must have reached N at some point during the block, and absorption has occurred by the end of it. That event has probability $p^N \geq m^N > 0$ and the blocks are independent. Hence $\mathbb{P}(\tau > kN) \leq (1 - m^N)^k$ for every integer $k \geq 0$. The bound is geometric in k , so $\mathbb{P}(\tau = \infty) = 0$ and, summing the tail, $\mathbb{E}[\tau] \leq N \sum_{k \geq 0} (1 - m^N)^k = Nm^{-N} < \infty$. $\square$

The bound in the proof is extremely crude and is not intended as an estimate of the duration; Section 4 computes $\mathbb{E}[\tau]$ exactly. Its purpose is only to establish finiteness, which is what the optional stopping argument will require.

Remark 2.2. Two conventions are worth fixing. First, the barriers are absorbing rather than reflecting, so the gambler cannot borrow and does not continue past the target. Section reflimitations discusses what changes when a barrier reflects. Second, the stake is exactly one unit on every round regardless of current capital. This is the assumption that the empirical section of this report subjects to the most scrutiny, because it is the one that real markets violate most flagrantly.

3 Ruin probability by difference equations

3.1 First-step analysis and the recurrence

Fix N and p and regard P_i as a function of the starting capital i . Condition on the outcome of the first round. From state i with $1 \leq i \leq N - 1$, the walk moves to $i + 1$ with probability p and to $i - 1$ with probability q , and by the Markov property the conditional probability of eventual success from the new state is P_{i+1} or P_{i-1} respectively, with no dependence on how state i was reached. The law of total probability therefore gives

$$P_i = p P_{i+1} + q P_{i-1}, \quad 1 \leq i \leq N - 1, \quad (5)$$

subject to the boundary conditions

$$P_0 = 0, \quad P_N = 1, \quad (6)$$

which record that the barriers are absorbing: from state 0 the target is never reached, and from state N it has been reached already. Equation (5) is a linear homogeneous second-order difference equation with constant coefficients, and (6) is a two-point boundary condition, so the solution is determined uniquely.

3.2 The characteristic equation

Seek solutions of (5) of the form $P_i = r^i$ for some $r \neq 0$. Substituting and dividing through by r^{i-1} , which is permissible since $r \neq 0$, gives

$$r^i = p r^{i+1} + q r^{i-1} \implies r = p r^2 + q \implies p r^2 - r + q = 0,$$

the characteristic equation. Its discriminant simplifies in a way that is worth displaying, because it is what makes the roots rational rather than merely real. Using $q = 1 - p$,

$$1 - 4pq = 1 - 4p(1 - p) = 1 - 4p + 4p^2 = (1 - 2p)^2 = (q - p)^2,$$

so the discriminant is a perfect square and

$$r = \frac{1 \pm \sqrt{1 - 4pq}}{2p} = \frac{1 \pm (q - p)}{2p}.$$

Taking the two signs in turn and using $p + q = 1$ throughout,

$$\begin{aligned} r_+ &= \frac{1 + (q - p)}{2p} = \frac{(p + q) + q - p}{2p} = \frac{2q}{2p} = \frac{q}{p}, \\ r_- &= \frac{1 - (q - p)}{2p} = \frac{(p + q) - q + p}{2p} = \frac{2p}{2p} = 1. \end{aligned}$$

The roots are therefore exactly $r = 1$ and $r = q/p$, as claimed. They are distinct precisely when $q \neq p$, that is when $p \neq 1/2$, and they coalesce at $r = 1$ when the game is fair. The two cases must consequently be handled separately, and it is the coalescence of the roots, not any peculiarity of the probability model, that is responsible for the different functional form in the symmetric case.

3.3 The asymmetric case

Proposition 3.1. *Let $p \neq 1/2$ and write $r = q/p$. Then for $0 \leq i \leq N$,*

$$P_i = \frac{1 - r^i}{1 - r^N} = \frac{1 - (q/p)^i}{1 - (q/p)^N}.$$

Proof. The roots 1 and $r = q/p$ are distinct, so the general solution of eqrefeq:recurrence is the linear combination

$$P_i = A \cdot 1^i + B r^i = A + B r^i, \quad A, B \in \mathbb{R}.$$

Imposing $P_0 = 0$ gives $A + B = 0$, hence $B = -A$ and $P_i = A(1 - r^i)$. Imposing $P_N = 1$ gives $A(1 - r^N) = 1$. Since $p \neq 1/2$ we have $r \neq 1$ and therefore $1 - r^N \neq 0$, so $A = (1 - r^N)^{-1}$ is well defined and

$$P_i = \frac{1 - r^i}{1 - r^N}.$$

Uniqueness follows because a second-order linear difference equation with two boundary conditions has a unique solution: the difference of two solutions satisfies the homogeneous equation with zero boundary data, and the recurrence propagates that zero across the whole range. $\square$

It is worth pausing on the structure of the answer. The quantity $r = q/p$ exceeds one exactly when the game is unfavourable. For $r > 1$ the numerator and denominator are both negative and the ratio is dominated by r^N , so P_i decays roughly like r^{i-N} : the probability of success falls off geometrically in the distance $N - i$ still to be travelled. For $r < 1$ the expression is close to $1 - r^i$ once N is moderately large, which is the infinite-barrier result anticipated in Section 7.

3.4 The symmetric case

Proposition 3.2. *Let $p = 1/2$. Then $P_i = i/N$ for $0 \leq i \leq N$.*

Proof. When $p = q = 1/2$ the characteristic equation becomes $\frac{1}{2}r^2 - r + \frac{1}{2} = \frac{1}{2}(r - 1)^2 = 0$, which has the single root $r = 1$ with multiplicity two. For a repeated root r of a linear difference equation the two linearly independent solutions are r^i and $i r^i$, here 1 and i , so the general solution is $P_i = A + Bi$. That these are solutions can be checked directly: with $p = q = 1/2$ the recurrence eqrefeq:recurrence reads $P_i = \frac{1}{2}P_{i+1} + \frac{1}{2}P_{i-1}$, equivalently

$$P_{i+1} - 2P_i + P_{i-1} = 0,$$

which states that the second difference of the sequence vanishes, and the sequences with vanishing second difference are precisely the affine ones. Imposing $P_0 = 0$ gives $A = 0$; imposing $P_N = 1$ gives $BN = 1$, so $B = 1/N$ and $P_i = i/N$. $\square$

The symmetric answer has an interpretation that makes it memorable. In a fair game the capital is a martingale, so the expected final capital equals the initial capital i ; since the final capital is N with probability P_i and 0 otherwise, $NP_i = i$. Section 5 turns this observation into a proof.

3.5 The symmetric case as a limit

Propositions 3.1 and 3.2 were obtained by separate arguments, and it is natural to ask whether the second is the limit of the first. It is, but the limit is genuinely indeterminate and deserves to be evaluated rather than asserted: as $p \rightarrow 1/2$ the ratio $r = q/p \rightarrow 1$, and both the numerator $1 - r^i$ and the denominator $1 - r^N$ tend to zero.

Proposition 3.3. *For fixed i and N with $0 \leq i \leq N$,*

$$\lim_{p \rightarrow 1/2} \frac{1 - (q/p)^i}{1 - (q/p)^N} = \frac{i}{N}.$$

Proof. The map $p \mapsto r = (1 - p)/p$ is continuous on $(0, 1)$ with $r \rightarrow 1$ as $p \rightarrow 1/2$, so it suffices to evaluate $g(r) = (1 - r^i)/(1 - r^N)$ as $r \rightarrow 1$. Both $1 - r^i \rightarrow 0$ and $1 - r^N \rightarrow 0$, so the quotient is of the indeterminate form $0/0$ and L'Hopital's rule applies. Differentiating numerator and denominator with respect to r ,

$$\frac{d}{dr} (1 - r^i) = -i r^{i-1}, \quad \frac{d}{dr} (1 - r^N) = -N r^{N-1},$$

and the derivative of the denominator is non-zero in a punctured neighbourhood of $r = 1$ because $N \geq 2$. Hence

$$\lim_{r \rightarrow 1} \frac{1 - r^i}{1 - r^N} = \lim_{r \rightarrow 1} \frac{-i r^{i-1}}{-N r^{N-1}} = \frac{i}{N},$$

and composing with $p \mapsto r$ gives the claim. Equivalently, without L'Hopital, factor both sides using $1 - r^k = (1 - r)(1 + r + \dots + r^{k-1})$ and cancel the common factor $(1 - r)$, which is legitimate for $r \neq 1$, to obtain

$$\frac{1 - r^i}{1 - r^N} = \frac{1 + r + \dots + r^{i-1}}{1 + r + \dots + r^{N-1}} \xrightarrow{r \rightarrow 1} \frac{i}{N},$$

since the numerator has i terms and the denominator N terms, each tending to one. The second argument is the more informative of the two, because the cancelled factor $(1 - r)$ is precisely the source of the indeterminacy. $\square$

The limit statement can be sharpened into a first-order sensitivity result, which is the quantitatively useful form and is used again in Section 7. Write $\varepsilon = p - \frac{1}{2}$, so that $p = \frac{1}{2} + \varepsilon$ and $q = \frac{1}{2} - \varepsilon$.

Proposition 3.4. *As $\varepsilon = p - \frac{1}{2} \rightarrow 0$, with i and N fixed,*

$$P_i = \frac{i}{N} + 2\varepsilon \frac{i(N-i)}{N} + O(\varepsilon^2).$$

Proof. Put $\theta = \log r = \log(1 - 2\varepsilon) - \log(1 + 2\varepsilon)$. Expanding each logarithm and using $\log(1 - x) - \log(1 + x) = -2(x + x^3/3 + \dots)$ with $x = 2\varepsilon$,

$$\theta = -4\varepsilon - \frac{16}{3}\varepsilon^3 + O(\varepsilon^5) = -4\varepsilon + O(\varepsilon^3).$$

Now $1 - r^k = 1 - e^{k\theta} = -(k\theta + \frac{1}{2}k^2\theta^2 + O(\theta^3))$, so

$$P_i = \frac{i\theta + \frac{1}{2}i^2\theta^2 + O(\theta^3)}{N\theta + \frac{1}{2}N^2\theta^2 + O(\theta^3)} = \frac{i}{N} \cdot \frac{1 + \frac{1}{2}i\theta + O(\theta^2)}{1 + \frac{1}{2}N\theta + O(\theta^2)} = \frac{i}{N} (1 + \frac{1}{2}(i - N)\theta + O(\theta^2)).$$

Substituting $\theta = -4\varepsilon + O(\varepsilon^3)$ gives

$$P_i = \frac{i}{N} + \frac{i(i - N)}{2N} (-4\varepsilon) + O(\varepsilon^2) = \frac{i}{N} + 2\varepsilon \frac{i(N - i)}{N} + O(\varepsilon^2),$$

as required. $\square$

The coefficient deserves comment. The quantity $i(N - i)$ appearing in Proposition 3.4 is exactly the expected duration of the fair game, derived independently in Section 4. The sensitivity of the ruin probability to a small edge is therefore proportional to how long the fair game would have lasted, which is the correct intuition: an edge has more opportunity to express itself the longer the play continues. Table 2 evaluates the exact asymmetric formula against both the symmetric limit and the first-order correction, confirming the rate.

Table 2: Convergence of the asymmetric ruin formula to the symmetric limit for $i = 30$, $N = 100$. The third column shows the deviation from i/N shrinking linearly in *varepsilon*; the fourth shows that deviation divided by the first-order prediction $2\varepsilon i(N - i)/N = 42$, which tends to one; the fifth shows the residual after the first-order term is removed, which shrinks quadratically.

ε	Exact P_i	$P_i - i/N$	Ratio to $2\varepsilon i(N - i)/N$	Residual after correction
10^{-1}	0.999994785	7.000e-01	0.16667	-3.500e+00
10^{-2}	0.711885674	4.119e-01	0.98068	-8.114e-03
10^{-3}	0.342998054	4.300e-02	1.02376	9.981e-04
10^{-4}	0.304211082	4.211e-03	1.00264	1.108e-05
10^{-5}	0.300420112	4.201e-04	1.00027	1.119e-07
10^{-6}	0.300042001	4.200e-05	1.00003	1.120e-09
10^{-7}	0.300004200	4.200e-06	1.00000	1.120e-11

The fourth column of Table 2 tends to one and the fifth falls by two orders of magnitude for each order of magnitude in ε , which is the signature of an $O(\varepsilon^2)$ residual and confirms Proposition 3.4 numerically.

3.6 A seventeenth century check

Proposition 3.1 specialises neatly when the two players start level. Setting $i = k$ and $N = 2k$ and dividing numerator and denominator by r^k ,

$$P_k = \frac{1 - r^k}{1 - r^{2k}} = \frac{1}{1 + r^k} = \frac{p^k}{p^k + q^k}, \quad (7)$$

so the odds in favour of the first player are exactly $p^k : q^k$. Huygens' fifth problem is this case. With three dice there are 27 ways to throw eleven and 15 ways to throw fourteen, and since all other throws are disregarded the conditional probability that a decisive throw favours A is $p = 27/42 = 9/14$. With $k = 12$ counters each, (7) gives odds of $9^{12} : 5^{12}$, that is **282,429,536,481** to **244,140,625**, and a probability of **0.999136** that A takes every counter. These are the numbers Huygens published in 1657, recovered here from the general formula with no intermediate rounding.

4 Expected duration of play

4.1 The inhomogeneous recurrence

The same first-step decomposition applied to the duration yields an inhomogeneous equation. From state i the walk uses one round to move, and from wherever it lands the expected remaining time is D_{i+1} or D_{i-1} . Taking expectations,

$$D_i = 1 + p D_{i+1} + q D_{i-1}, \quad 1 \leq i \leq N - 1, \quad (8)$$

with boundary conditions

$$D_0 = D_N = 0, \quad (9)$$

since play has already ended at either barrier. Lemma 2.1 guarantees that the quantities D_i are finite, so the manipulation is legitimate; without that guarantee (8) would be a relation among possibly infinite quantities.

The homogeneous part of (8) is exactly (5), so its general solution is already known: $A + B(q/p)^i$ when $p \neq 1/2$ and $A + Bi$ when $p = 1/2$. What remains is a particular solution, and the standard method of undetermined coefficients requires care because the forcing term, the constant 1, is annihilated by the homogeneous operator: the constant sequence is itself a homogeneous solution, since $r = 1$ is always a root. The trial polynomial must therefore be raised in degree, once in the asymmetric case where $r = 1$ is a simple root, and twice in the symmetric case where it is a double root.

4.2 The asymmetric case

Proposition 4.1. *Let $p \neq 1/2$. Then for $0 \leq i \leq N$,*

$$D_i = \frac{i}{q-p} - \frac{N}{q-p} \cdot \frac{1 - (q/p)^i}{1 - (q/p)^N} = \frac{i}{q-p} - \frac{N}{q-p} P_i.$$

Proof. Since $r = 1$ is a simple root of the characteristic polynomial, try the first-degree particular solution $D_i^* = ci$ for a constant c to be determined. Substituting into [eqrefeq:durrec](#),

$$ci = 1 + pc(i+1) + qc(i-1) = 1 + ci(p+q) + c(p-q) = 1 + ci + c(p-q),$$

where $p + q = 1$ was used. Cancelling ci from both sides leaves $0 = 1 + c(p - q)$, so

$$c = \frac{1}{q - p},$$

which is finite precisely because $p \neq q$. Hence $D_i^* = i/(q - p)$ and the general solution of $\text{eqref{eq:durrec}}$ is

$$D_i = \frac{i}{q - p} + A + B \left(\frac{q}{p} \right)^i.$$

Write $r = q/p$. The boundary condition $D_0 = 0$ gives $A + B = 0$, so $A = -B$ and

$$D_i = \frac{i}{q - p} + B(r^i - 1).$$

The boundary condition $D_N = 0$ gives

$$\frac{N}{q - p} + B(r^N - 1) = 0 \implies B = \frac{-N}{(q - p)(r^N - 1)} = \frac{N}{(q - p)(1 - r^N)}.$$

Substituting back,

$$D_i = \frac{i}{q - p} + \frac{N(r^i - 1)}{(q - p)(1 - r^N)} = \frac{i}{q - p} - \frac{N}{q - p} \cdot \frac{1 - r^i}{1 - r^N},$$

which is the stated formula, the final factor being P_i by Proposition 3.1. $\square$

The compact form $D_i = (i - NP_i)/(q - p)$ is the one used in the code accompanying this report, both because it reuses an already tested routine and because it is better conditioned numerically than the fully expanded expression when q/p is close to one. Section 5 shows that this form is not merely an algebraic rearrangement but is exactly what Wald's identity delivers.

4.3 The symmetric case

Proposition 4.2. *Let $p = 1/2$. Then $D_i = i(N - i)$ for $0 \leq i \leq N$.*

Proof. Here $r = 1$ is a double root, so both 1 and i solve the homogeneous equation and the particular solution must be sought at second degree. Try $D_i^* = ci^2$. Substituting into $\text{eqref{eq:durrec}}$ with $p = q = 1/2$,

$$ci^2 = 1 + \frac{1}{2}c(i + 1)^2 + \frac{1}{2}c(i - 1)^2 = 1 + \frac{1}{2}c[(i^2 + 2i + 1) + (i^2 - 2i + 1)] = 1 + ci^2 + c.$$

The cross terms cancel, leaving $0 = 1 + c$, so $c = -1$ and $D_i^* = -i^2$. The general solution is therefore $D_i = -i^2 + A + Bi$. The condition $D_0 = 0$ gives $A = 0$, and $D_N = 0$ gives $-N^2 + BN = 0$, so $B = N$. Hence

$$D_i = -i^2 + Ni = i(N - i).$$

$\square$

The symmetric result is striking in its own right. A gambler starting with half of a target of $N = 100$ units expects to play 2,500 rounds before the game ends, even though the outcome is a coin flip. The expected duration is the product of the two players' stakes, a fact Huygens' contemporaries would have found surprising and which remains the cleanest illustration of how slowly a fair random walk covers ground: displacement grows like the square root of time, so covering a distance of order N takes a time of order N^2 .

Corollary 4.3. *The symmetric duration is the limit of the asymmetric one: $\lim_{p \rightarrow 1/2} D_i = i(N - i)$.*

Proof. Write $\varepsilon = p - \frac{1}{2}$, so $q - p = -2\varepsilon$. By Proposition 3.4, $P_i = i/N + 2\varepsilon i(N-i)/N + O(\varepsilon^2)$, whence

$$i - NP_i = i - N \left(\frac{i}{N} + 2\varepsilon \frac{i(N-i)}{N} \right) + O(\varepsilon^2) = -2\varepsilon i(N-i) + O(\varepsilon^2).$$

Dividing by $q - p = -2\varepsilon$,

$$D_i = \frac{i - NP_i}{q - p} = \frac{-2\varepsilon i(N-i) + O(\varepsilon^2)}{-2\varepsilon} = i(N-i) + O(\varepsilon) \xrightarrow{\varepsilon \rightarrow 0} i(N-i).$$

□

Corollary 4.3 also explains a numerical caveat. The expression $(i - NP_i)/(q - p)$ evaluates a ratio whose numerator and denominator both vanish as $p \rightarrow 1/2$, so in floating point it loses roughly one significant digit for each order of magnitude by which $|\varepsilon|$ falls below one. The implementation accordingly switches to the exact symmetric formula when $|p - 1/2| < 10^{-10}$, and the report notes this rather than leaving it implicit.

5 An independent derivation by martingales

The derivations of Section 3 rest entirely on manipulating a difference equation, and an error in that manipulation would propagate silently into every later result. This section derives the ruin probability a second time by a route that shares no algebra with the first, using the optional stopping theorem. Agreement between two independent derivations is the check on the work; it is reported as such rather than presented as a stylistic variation.

Let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ be the natural filtration of the increments, so that S_n is $\mathcal{F}_n$ measurable and X_{n+1} is independent of $\mathcal{F}_n$. Recall Doob's optional stopping theorem (Doob 1953; Williams 1991): if (M_n) is a martingale, τ a stopping time, and either τ is bounded, or τ is almost surely finite and $(M_{n \wedge \tau})$ is uniformly bounded, or $\mathbb{E}[\tau] < \infty$ and the increments of M are bounded, then $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$. In every application below the walk is confined to $\{0, 1, \dots, N\}$ before absorption and τ is almost surely finite by Lemma 2.1, so the second condition holds and the theorem applies. This verification is the part of the argument that is most often omitted, and it is not a formality: the theorem fails without some such condition, as the unbounded simple random walk on $\mathbb{Z}$ shows.

5.1 The fair game

Proposition 5.1. *Let $p = 1/2$. Then (S_n) is a martingale and $P_i = i/N$.*

Proof. The walk is bounded before absorption and hence integrable. For the martingale property, condition on $\mathcal{F}_n$ and use independence of X_{n+1} :

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[X_{n+1}] = S_n + (p - q) = S_n,$$

since $p = q = 1/2$ gives $\mathbb{E}[X_{n+1}] = 0$. Thus (S_n) is a martingale with $\mathbb{E}[S_0] = i$.

The stopped process $S_{n \wedge \tau}$ takes values in $\{0, 1, \dots, N\}$ and is therefore uniformly bounded, and $\tau < \infty$ almost surely by Lemma 2.1. Optional stopping gives $\mathbb{E}[S_\tau] = \mathbb{E}[S_0] = i$. But S_τ takes only the values 0 and N , with $\mathbb{P}(S_\tau = N) = P_i$, so

$$\mathbb{E}[S_\tau] = 0 \cdot (1 - P_i) + N \cdot P_i = NP_i.$$

Equating the two expressions for $\mathbb{E}[S_\tau]$ gives $NP_i = i$, so $P_i = i/N$, in agreement with Proposition 3.2. □

The same device delivers the expected duration in the fair case, which is a useful independent check on Proposition 4.2.

Proposition 5.2. *Let $p = 1/2$. Then $M_n = S_n^2 - n$ is a martingale, and $D_i = i(N - i)$.*

Proof. Since $X_{n+1}^2 = 1$ identically and $\mathbb{E}[X_{n+1}] = 0$,

$$\mathbb{E}[S_{n+1}^2 \mid \mathcal{F}_n] = \mathbb{E}[(S_n + X_{n+1})^2 \mid \mathcal{F}_n] = S_n^2 + 2S_n\mathbb{E}[X_{n+1}] + \mathbb{E}[X_{n+1}^2] = S_n^2 + 1,$$

so $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = S_n^2 + 1 - (n + 1) = M_n$ and (M_n) is a martingale with $\mathbb{E}[M_0] = i^2$. The increments of M are bounded on $\{0, \dots, N\}$ and $\mathbb{E}[\tau] < \infty$ by Lemma 2.1, so optional stopping applies and $\mathbb{E}[S_\tau^2 - \tau] = i^2$. Since S_τ^2 equals N^2 with probability $P_i = i/N$ and 0 otherwise,

$$\mathbb{E}[\tau] = \mathbb{E}[S_\tau^2] - i^2 = N^2 \cdot \frac{i}{N} - i^2 = iN - i^2 = i(N - i),$$

which is Proposition 4.2. □

5.2 The unfair game and the exponential martingale

When $p \neq 1/2$ the capital itself is no longer a martingale, since $\mathbb{E}[X_{n+1}] = p - q \neq 0$. The standard remedy is to find a function of the capital that is a martingale, and for a walk with independent increments the natural candidate is an exponential.

Proposition 5.3. *Let $p \neq 1/2$ and set $r = q/p$. Then $M_n = r^{S_n}$ is a martingale, and*

$$P_i = \frac{1 - r^i}{1 - r^N}.$$

Proof. The process is bounded before absorption, since $S_n \in \{0, \dots, N\}$ and hence r^{S_n} lies between $\min(1, r^N)$ and $\max(1, r^N)$; in particular it is integrable. For the martingale property, condition on $\mathcal{F}_n$ and sum over the two possible increments:

$$\mathbb{E}[r^{S_{n+1}} \mid \mathcal{F}_n] = pr^{S_n+1} + qr^{S_n-1} = r^{S_n} \left(pr + \frac{q}{r} \right).$$

Substituting $r = q/p$,

$$pr + \frac{q}{r} = p \cdot \frac{q}{p} + q \cdot \frac{p}{q} = q + p = 1,$$

so $\mathbb{E}[r^{S_{n+1}} \mid \mathcal{F}_n] = r^{S_n} = M_n$, and (M_n) is a martingale with $\mathbb{E}[M_0] = r^i$. The choice $r = q/p$ is not arbitrary: it is the non-unit root of $pr^2 - r + q = 0$ found in Section [refruin-prob](#), which is exactly the condition $pr + q/r = 1$ after multiplication by r . The two derivations meet at the characteristic polynomial even though they proceed differently from there.

The stopped process $M_{n \wedge \tau}$ is uniformly bounded and τ is almost surely finite by Lemma 2.1, so optional stopping gives $\mathbb{E}[r^{S_\tau}] = r^i$. Since $S_\tau \in \{0, N\}$ with $\mathbb{P}(S_\tau = N) = P_i$,

$$(1 - P_i)r^0 + P_i r^N = r^i \implies 1 + P_i(r^N - 1) = r^i \implies P_i = \frac{r^i - 1}{r^N - 1} = \frac{1 - r^i}{1 - r^N},$$

the division being legitimate because $r \neq 1$ implies $r^N \neq 1$. This is precisely Proposition 3.1. □

Finally, Wald's identity recovers the asymmetric duration and explains the compact form noted after Proposition 4.1.

Proposition 5.4. *Let $p \neq 1/2$. Then $M_n = S_n - n(p - q)$ is a martingale and $D_i = (i - NP_i)/(q - p)$.*

Proof. Since $\mathbb{E}[X_{n+1}] = p - q$, $\mathbb{E}[S_{n+1} - (n+1)(p - q) \mid \mathcal{F}_n] = S_n + (p - q) - (n+1)(p - q) = M_n$, so (M_n) is a martingale with $\mathbb{E}[M_0] = i$. The increments are bounded and $\mathbb{E}[\tau] < \infty$, so optional stopping gives $\mathbb{E}[S_\tau] - (p - q)\mathbb{E}[\tau] = i$. With $\mathbb{E}[S_\tau] = NP_i$ and $\mathbb{E}[\tau] = D_i$ this rearranges to

$$D_i = \frac{NP_i - i}{p - q} = \frac{i - NP_i}{q - p},$$

which is the form used in the implementation and, on substituting Proposition 3.1 for P_i , is identical to Proposition 4.1. $\square$

5.3 Numerical confirmation that the two derivations agree

The martingale route produces an identity that the difference-equation route does not obviously satisfy, namely $\mathbb{E}[r^{S_\tau}] = r^i$. Checking it numerically tests the two derivations against one another.

Across the 24 parameter combinations tested, the largest relative discrepancy between the two derivations is 7.87×10^{-15} . That is at the level of double precision rounding, so the two derivations agree to the limit of what the arithmetic can resolve. The comparison is recorded as a verification check with an explicit tolerance, so the agreement is enforced rather than merely asserted.

One implementation detail is worth recording, because it was discovered by this check failing. The probability of absorption at the origin is mathematically $1 - P_i$, but evaluating it that way is numerically destructive when P_i is close to one: for $i = 45$, $N = 100$ and $p = 0.61$ the true value is approximately 1.812×10^{-9} , and forming it as $1 - P_i$ in double precision leaves only about eight correct significant digits because the subtraction cancels the leading ones. The library therefore evaluates

$$1 - P_i = \frac{r^i - r^N}{1 - r^N}$$

directly on the log scale rather than by subtraction, which preserves full relative precision. Every ruin probability reported in this document uses that stable form. The episode is a small illustration of a general point: an identity that is exact in real arithmetic can still be evaluated badly, and the way to find out is to check it against an independent route and watch the check fail.

6 Matrix methods and numerical cross-validation

A third and entirely numerical route to the same quantities runs through the canonical decomposition of the absorbing chain (Kemeny and Snell 1976). It shares neither the algebra of Section 3 nor the probabilistic reasoning of Section 5, and so provides an independent check on both.

6.1 Canonical form and the fundamental matrix

Order the state space with the transient states first and the absorbing states last, that is $(1, 2, \dots, N-1, 0, N)$. In this ordering the transition matrix (4) takes the block form

$$P = \begin{pmatrix} Q & R \\ 0 & I_2 \end{pmatrix}, \tag{10}$$

where Q is the $(N-1) \times (N-1)$ substochastic matrix of transitions among transient states, R is the $(N-1) \times 2$ matrix of transitions from transient to absorbing states, and I_2 is the identity of order two. Here $Q_{j,j+1} = p$ and $Q_{j,j-1} = q$ for interior j , while R has q in its first row and first column, p in its last row and second column, and zeros elsewhere.

Because every transient state leads to absorption with positive probability, $Q^n \rightarrow 0$ as $n \rightarrow \infty$, the matrix $I - Q$ is invertible, and the Neumann series converges:

$$M = (I - Q)^{-1} = \sum_{n=0}^{\infty} Q^n. \quad (11)$$

M is the fundamental matrix of the absorbing chain, and M_{jk} has a direct reading: since $(Q^n)_{jk}$ is the probability of being in transient state k at time n having started in j , the sum M_{jk} is the expected number of visits to k before absorption starting from j . Two standard consequences follow immediately. The vector of expected absorption times is

$$\mathbf{t} = M\mathbf{1}, \quad t_j = D_j, \quad (12)$$

because the total time to absorption is the total number of visits to transient states, and the matrix of absorption probabilities is

$$B = MR, \quad B_{j,2} = P_j, \quad (13)$$

the second column corresponding to absorption at N .

6.2 Cross-validation

Table 3 compares the closed forms of Sections 3 and 4 with the values obtained by forming $(I - Q)^{-1}$ numerically and applying (12) and (13). The matrix computation involves no formula from those sections; it inverts a matrix.

Table 3: Closed-form results compared with the fundamental matrix computation. The two routes share no code path: the left-hand values come from the algebraic solutions of the difference equations, the right-hand values from numerically inverting $I - Q$. Agreement is at the level of floating-point rounding throughout.

i	N	p	P_i closed form	P_i matrix	abs. diff.	D_i closed form	D_i matrix	rel. diff.
10	80	0.44	0.00000004	0.00000004	1.4e-21	83.333	83.333	5.5e-15
25	80	0.44	0.00000173	0.00000173	4.6e-20	208.332	208.332	8.5e-15
60	80	0.44	0.00804068	0.00804068	7.1e-17	494.640	494.640	1.8e-14
10	120	0.44	0.00000000	0.00000000	1.5e-25	83.333	83.333	5.8e-15
25	120	0.44	0.00000000	0.00000000	5.2e-24	208.333	208.333	8.7e-15
60	120	0.44	0.00000052	0.00000052	1.5e-20	499.999	499.999	1.8e-14
10	80	0.50	0.12500000	0.12500000	1.2e-15	700.000	700.000	8.9e-15
25	80	0.50	0.31250000	0.31250000	2.4e-15	1,375.000	1,375.000	9.4e-15
60	80	0.50	0.75000000	0.75000000	1.0e-15	1,200.000	1,200.000	2.1e-15
10	120	0.50	0.08333333	0.08333333	1.5e-15	1,100.000	1,100.000	1.2e-14
25	120	0.50	0.20833333	0.20833333	3.6e-15	2,375.000	2,375.000	1.3e-14
60	120	0.50	0.50000000	0.50000000	4.1e-15	3,600.000	3,600.000	3.3e-15
10	80	0.57	0.94030527	0.94030527	1.1e-16	465.889	465.889	0.0e+00
25	80	0.57	0.99912936	0.99912936	1.4e-15	392.360	392.360	1.7e-15
60	80	0.57	0.99999995	0.99999995	6.7e-16	142.857	142.857	0.0e+00
10	120	0.57	0.94030527	0.94030527	2.0e-15	734.547	734.547	1.5e-16
25	120	0.57	0.99912936	0.99912936	1.2e-15	677.825	677.825	6.7e-16
60	120	0.57	0.99999995	0.99999995	2.7e-15	428.571	428.571	1.6e-15

The largest absolute discrepancy in the ruin probabilities is 4.05×10^{-15} and the largest relative discrepancy in the expected durations is 1.77×10^{-14} . Both are attributable to floating-point rounding rather than to any difference in the mathematics.

There is a further structural check available. The fundamental matrix should reproduce the expected number of visits to each state, and summing that row should return the expected duration. The interior row sums of M are compared with D_i implicitly by (12), but the diagonal carries additional information: M_{ii} is the expected number of visits to the starting state itself, which for a fair walk with $N = 40$ and $i = 20$ is **20.000** visits.

7 Asymptotics and limiting behaviour

The closed forms invite three questions. What happens when the opponent's capital is effectively unlimited, how sensitive is the answer to a small departure from fairness, and how do those two effects interact as the target recedes. This section answers each in turn, and the answers together make the case that the gambler's ruin problem is less a curiosity than a statement about the survival of any finite bankroll facing a persistent edge.

7.1 An infinitely rich opponent

Theorem 7.1. *Fix the initial capital $i \geq 1$ and let $N \rightarrow \infty$. Write $\rho_i = \lim_{N \rightarrow \infty} \mathbb{P}(\text{ruin} \mid S_0 = i)$. Then*

$$\rho_i = \begin{cases} 1, & p \leq \frac{1}{2}, \\ \left(\frac{q}{p}\right)^i, & p > \frac{1}{2}. \end{cases}$$

Proof. Consider the three cases separately.

If $p < 1/2$ then $r = q/p > 1$, so $r^N \rightarrow \infty$. From Proposition 3.1,

$$P_i = \frac{1 - r^i}{1 - r^N} \rightarrow 0 \quad (N \rightarrow \infty),$$

because the numerator is a fixed finite number and the denominator diverges to $-\infty$. Hence $\rho_i = 1 - P_i \rightarrow 1$.

If $p = 1/2$ then $P_i = i/N \rightarrow 0$ by Proposition 3.2, so again $\rho_i = 1$.

If $p > 1/2$ then $r = q/p < 1$, so $r^N \rightarrow 0$ and

$$P_i = \frac{1 - r^i}{1 - r^N} \rightarrow 1 - r^i,$$

whence $\rho_i = 1 - P_i = r^i = (q/p)^i$.

Monotonicity in N justifies interpreting the limit as the ruin probability of the walk on $\{0, 1, 2, \dots\}$ with a single absorbing barrier at the origin: the events $\{\text{ruin before reaching } N\}$ increase to $\{\text{ruin}\}$ as N grows, and continuity from below of the probability measure gives the identification. $\square$

Theorem 7.1 is the formal version of the claim that a player of finite means facing an unlimited opponent is doomed. It is worth stating the consequence explicitly, because it is stronger than the intuition usually offered.

Corollary 7.2. *A gambler with any finite capital i , playing against an opponent of unlimited capital at odds that are fair or worse, is ruined with probability one. In particular the conclusion holds at $p = 1/2$, where the gambler has no disadvantage at all, and the expected duration until that certain ruin is infinite.*

Proof. The probability statement is the case $p \leq 1/2$ of Theorem 7.1. For the duration, take $p = 1/2$ and note $D_i = i(N - i) \rightarrow \infty$ as $N \rightarrow \infty$; ruin is certain but its expected waiting time is unbounded, so the walk

is null recurrent rather than positive recurrent. For $p < 1/2$, Proposition 4.1 gives $D_i \rightarrow i/(q - p)$, which is finite: an adverse drift brings ruin not only surely but quickly. $\square$

The contrast inside Corollary 7.2 is the interesting part. At exactly fair odds ruin is certain but arrives after an infinite expected wait; with even a slight adverse drift it is certain and arrives in expected time $i/(q - p)$, which for $p = 0.49$ and $i = 100$ units is only **5,000** rounds. The transition at $p = 1/2$ is therefore not a transition in whether the gambler is ruined but in how long the ruin takes.

Conversely, a favourable game never confers safety. Even with $p > 1/2$ the ruin probability $(q/p)^i$ is strictly positive for every finite i , though it decays geometrically in the bankroll. This is the quantitative content of the requirement that a strategy with positive expectation still needs capital to survive its own variance.

7.2 Sensitivity to a small edge

Proposition 3.4 gives the local behaviour near fairness: $P_i \approx i/N + 2\varepsilon i(N - i)/N$ where $\varepsilon = p - 1/2$. Two features of that expression matter in practice. The coefficient $i(N - i)/N$ is maximised at $i = N/2$ and grows linearly in N there, taking the value $N/4$. A given edge therefore has an effect on the ruin probability that grows in proportion to the size of the game, so the same house advantage that is nearly invisible over a short session becomes decisive over a long one. Figure 1 shows this directly: the fair game gives the straight line $P_i = i/N$, and even an edge of one percentage point bends it visibly away.

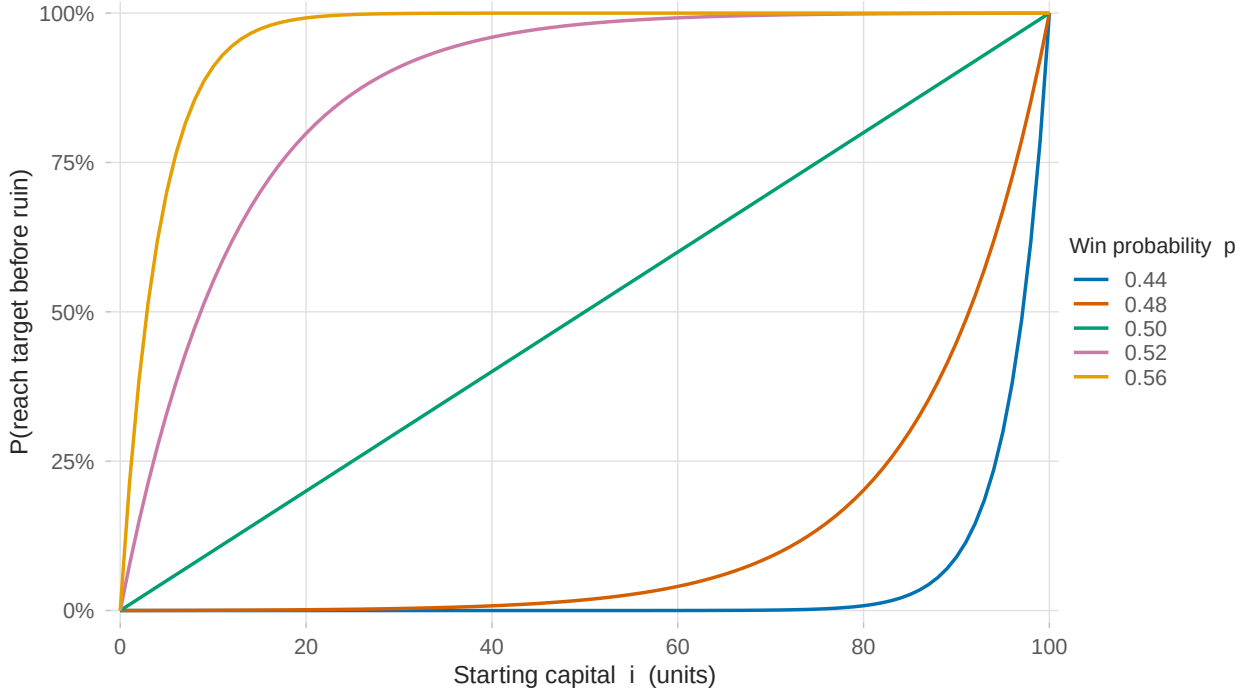


Figure 1: Probability of reaching the target before ruin, as a function of starting capital, for a target of $N = 100$ units. The fair game is the straight diagonal; small departures from fairness bend it sharply. At $p = 0.48$ a gambler starting with three quarters of the target still faces a modelled ruin probability of 86.5%, which is the practical force of a two percentage point disadvantage.

The second feature is the interaction with N . Holding the edge fixed and letting the target recede, the ruin probability approaches one geometrically whenever $p < 1/2$. Table 4 quantifies the degradation for a gambler who always starts with a quarter of the target.

Table 4: Modelled probability of ruin for a gambler starting with a quarter of the target, as the target grows. Reading along any row shows the cost of a small edge; reading down a column shows that the cost compounds with the size of the game. The final column is the fair game, where the ruin probability is the constant three quarters.

N	$p = 0.485$	$p = 0.490$	$p = 0.495$	$p = 0.500$
20	84.92%	81.94%	78.62%	75.00%
50	94.15%	89.85%	83.47%	75.00%
100	99.14%	96.80%	89.85%	75.00%
200	99.99%	99.79%	96.79%	75.00%
500	> 99.99%	> 99.99%	99.95%	75.00%
1000	> 99.99%	> 99.99%	> 99.99%	75.00%

The final column is constant at $1 - i/N = 3/4$, since in the fair game only the ratio of capital to target matters. Every other column rises toward certainty without ever reaching it: for a finite barrier the ruin probability is strictly below one, so the largest entries are shown as lower bounds rather than as an equality, several of them being indistinguishable from one in double precision. The lesson is that scale is not neutral: a disadvantaged gambler is harmed by playing a bigger game even when the ratio of bankroll to target is held fixed, which is the opposite of the fair-game case. Figure 2 presents the same information across the whole (p, i) plane.

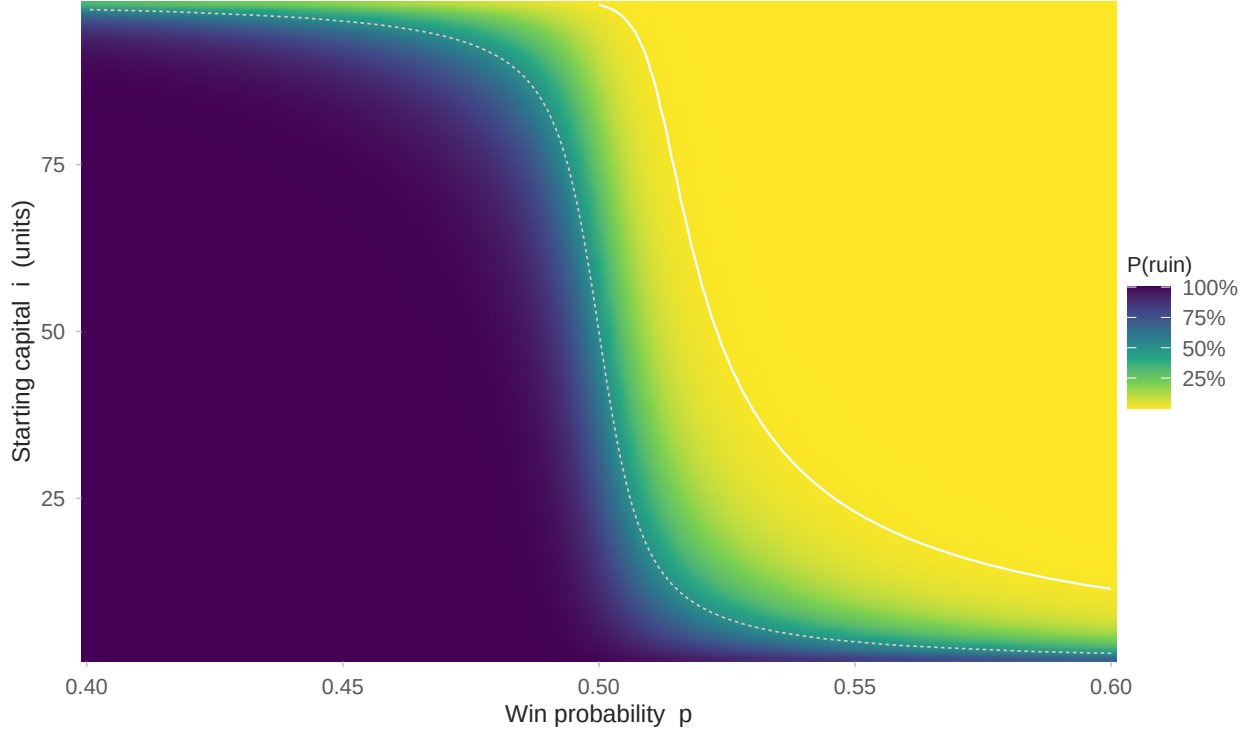


Figure 2: Modelled probability of ruin over the win probability and starting capital, for a fixed target of $N = 100$ units. The transition from near-certain ruin to near-certain success is compressed into a narrow band around $p = 0.5$, and that band narrows further as starting capital rises. The contour marks the 1% ruin level.

8 The diffusion limit

The discrete model has a continuous counterpart, and the correspondence is exact in the limit. This section states the scaling, quotes the invariance principle rather than proving it, and verifies that the continuous hitting probability is the limit of Proposition 3.1.

8.1 Scaling

Let the stake be $h > 0$ rather than one unit, let the gambler play once every $\Delta t = h^2/\sigma^2$ units of time for a fixed $\sigma > 0$, and let the win probability depend on the stake through

$$p_h = \frac{1}{2} \left(1 + \frac{\mu h}{\sigma^2} \right), \quad q_h = 1 - p_h, \quad (14)$$

for a fixed $\mu \in \mathbb{R}$ and all h small enough that $p_h \in (0, 1)$. Under this scaling a single round has mean and variance

$$\mathbb{E}[hX] = h(p_h - q_h) = h \cdot \frac{\mu h}{\sigma^2} = \mu \Delta t, \quad \text{Var}(hX) = h^2 (1 - (p_h - q_h)^2) = \sigma^2 \Delta t (1 + O(h^2)),$$

so that over any fixed time interval the accumulated mean is μ times the elapsed time and the accumulated variance is σ^2 times it, both independently of h . This is the correct balance: the stake shrinks like h while the number of rounds per unit time grows like h^{-2} , and the edge shrinks like h so that drift and diffusion remain comparable.

Theorem 8.1 (Donsker's invariance principle). *Let $S_t^{(h)}$ denote the piecewise constant interpolation of the scaled walk with stake h and win probability p_h , started at $S_0^{(h)} = x$. Then as $h \downarrow 0$, $S^{(h)}$ converges weakly in the Skorokhod space $D[0, \infty)$ to the diffusion*

$$X_t = x + \mu t + \sigma W_t,$$

Brownian motion with drift μ and volatility σ , where W is a standard Brownian motion.

The result is quoted rather than proved; see Donsker (1951) and, for the modern treatment in the Skorokhod topology, Billingsley (1999). The functional form of the convergence matters here, as opposed to convergence of the marginal distributions alone, because the quantity of interest is a hitting probability, a functional of the whole path. Hitting times of an interval are almost surely continuous functionals of the limit path, so the continuous mapping theorem transfers the convergence to the absorption probabilities.

8.2 Hitting probabilities for drifted Brownian motion

Proposition 8.2. *Let $X_t = x + \mu t + \sigma W_t$ with $\sigma > 0$ and let $T = \inf\{t \geq 0 : X_t \notin (0, L)\}$ for $0 < x < L$. Then*

$$\mathbb{P}(X_T = L) = \begin{cases} \frac{1 - e^{-2\mu x/\sigma^2}}{1 - e^{-2\mu L/\sigma^2}}, & \mu \neq 0, \\ \frac{x}{L}, & \mu = 0. \end{cases}$$

Proof. Let $u(x) = \mathbb{P}(X_T = L \mid X_0 = x)$. The function u is harmonic for the infinitesimal generator $\mathcal{A} = \frac{1}{2}\sigma^2 \partial_{xx} + \mu \partial_x$ of the diffusion, so it solves the boundary value problem

$$\frac{1}{2}\sigma^2 u''(x) + \mu u'(x) = 0, \quad u(0) = 0, \quad u(L) = 1,$$

which is the continuous analogue of the difference equation

eqref{eq:recurrence}: the generator replaces the one-step averaging operator, and the boundary conditions are identical. Writing $v = u'$ reduces the equation to $v' = -(2\mu/\sigma^2)v$, so $v(x) = C \exp(-2\mu x/\sigma^2)$ and, integrating, $u(x) = A + B \exp(-2\mu x/\sigma^2)$ for $\mu \neq 0$. Imposing $u(0) = 0$ gives $A = -B$, and imposing $u(L) = 1$ gives $B(e^{-2\mu L/\sigma^2} - 1) = 1$, whence the stated formula. When $\mu = 0$ the equation is $u'' = 0$, so u is affine and the boundary conditions force $u(x) = x/L$. $\square$

8.3 The discrete formula converges to the continuous one

The correspondence between Propositions 3.1 and 8.2 can be made explicit. Under the scaling (14), a capital of x in continuous units corresponds to $i = x/h$ discrete units and a barrier at L corresponds to $N = L/h$. The ratio appearing in the discrete formula is

$$\log\left(\frac{q_h}{p_h}\right) = \log\left(1 - \frac{\mu h}{\sigma^2}\right) - \log\left(1 + \frac{\mu h}{\sigma^2}\right) = -\frac{2\mu h}{\sigma^2} + O(h^3),$$

so that

$$\left(\frac{q_h}{p_h}\right)^i = \exp\left(\frac{x}{h} \left[-\frac{2\mu h}{\sigma^2} + O(h^3)\right]\right) = \exp\left(-\frac{2\mu x}{\sigma^2}\right) + O(h^2),$$

and identically with L in place of x . Substituting into Proposition 3.1 recovers Proposition 8.2 term by term. The discrete ruin formula is therefore not merely analogous to the diffusion result; it converges to it, and the exponential $e^{-2\mu x/\sigma^2}$ is the continuous shadow of the ratio $(q/p)^i$. Table 5 verifies the convergence numerically.

Table 5: Convergence of the discrete ruin probability to the drifted Brownian motion hitting probability, for $\mu = 0.30$, $\sigma = 1.0$, $x = 3$ and $L = 10$. As the stake h shrinks the discrete answer approaches the continuous one, with error falling roughly in proportion to h .

h	p_h	$i = x/h$	$N = L/h$	Discrete P_i	Diffusion limit	Absolute error
0.500	0.57500	6	20	0.83893464	0.83677527	2.16e-03
0.200	0.53000	15	50	0.83711863	0.83677527	3.43e-04
0.100	0.51500	30	100	0.83686104	0.83677527	8.58e-05
0.050	0.50750	60	200	0.83679671	0.83677527	2.14e-05
0.020	0.50300	150	500	0.83677870	0.83677527	3.43e-06
0.010	0.50150	300	1,000	0.83677613	0.83677527	8.57e-07

At the finest scaling tested, a stake of $h = 0.01$ units, the discrete and continuous answers differ by 8.57×10^{-7} , and the error falls monotonically as the stake shrinks.

A brief word on the wider significance. The generator $\mathcal{A}$ whose harmonic functions solve the hitting problem in Proposition 8.2 is the same operator that, applied to a derivative price under a geometric Brownian motion, produces the Black-Scholes equation (Black and Scholes 1973), and the exponential martingale $(q/p)^{S_n}$ used in Proposition 5.3 is the discrete counterpart of the density used to change to the risk-neutral measure. The gambler's ruin problem is thus the simplest non-trivial member of the family of boundary value problems that derivative pricing inhabits.

9 Monte Carlo study

Closed forms that have been derived twice and checked against a matrix computation are unlikely to be wrong, but all three routes share the assumption that the model has been specified correctly. Simulation tests something different: it samples the process itself, and agreement between the sampled process and the formula confirms that the formula describes the walk rather than some adjacent object.

9.1 Implementation

All simulation in this report runs through a single engine. Walks are advanced in blocks rather than one round at a time: for every still-active replication a block of increments is drawn at once, the running capital is formed with a vectorised cumulative sum, and the first crossing of either barrier is located by a column-wise maximum. Replications that are absorbed within the block leave the active set, and the block length doubles, so that the heavy right tail of the duration distribution costs a handful of iterations rather than thousands. No loop over individual rounds is executed anywhere. The same engine serves both the plus or minus one walk and the bootstrap resampling of Section 10, differing only in the function supplied to draw increments, which keeps the idealised and empirical experiments strictly comparable.

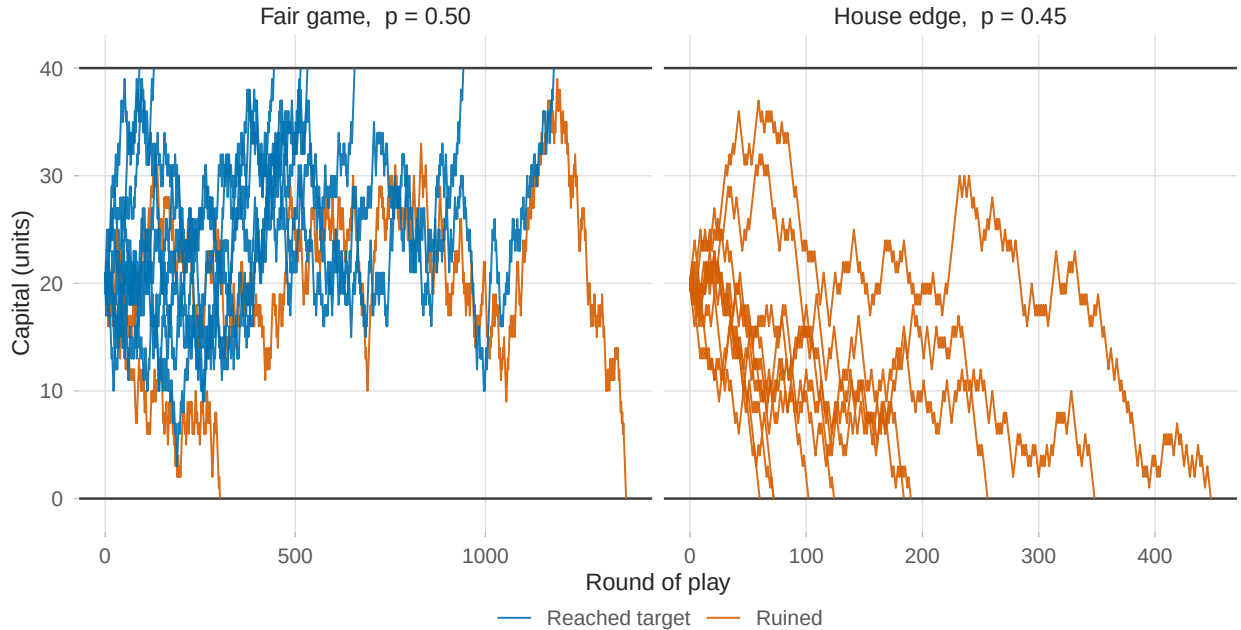


Figure 3: 10 simulated capital paths under a fair game and under a two-sided disadvantage, starting from 20 units with barriers at 0 and 40. Paths are coloured by which barrier absorbed them. The fair panel shows the characteristic aimless wandering that makes the duration so long; the disadvantaged panel shows the same wandering with a downward tilt that ends most paths at the lower barrier.

Figure 3 shows ten paths in each of two regimes. Under the fair game the paths wander without any systematic direction and take a long time to find a barrier; under a disadvantage of five percentage points the same wandering acquires a tilt, and 10 of the 10 paths shown end in ruin.

9.2 Estimating the ruin probability

The main experiment estimates P_i on a grid of parameter values with 100,000 independent replications in every cell. Each replication is a Bernoulli trial with success probability P_i , so the estimator is a sample

proportion and interval estimates follow from binomial theory. Two intervals are reported: the Wilson score interval (Wilson 1927), which has good coverage without the degeneracies of the Wald interval near zero and one, and the Clopper-Pearson interval (Clopper and Pearson 1934), which is conservative by construction. Brown et al. (2001) survey the trade-off and recommend Wilson for general use, which is why it is the interval quoted in the text.

Table 6: Monte Carlo estimates of the probability of reaching the target against the closed form, with 100,000 replications per cell and a target of $N = 100$ units. The standardised deviation z in the final column is the error in units of its own standard error; all nine values are within the range expected from sampling variation alone, and every interval covers the exact value.

i	p	Exact P_i	Monte Carlo	Abs. error	Wilson 95% interval	Clopper-Pearson 95%	z
25	0.48	0.002138	0.002250	1.12e-04	[0.001975, 0.002563]	[0.001966, 0.002564]	0.77
50	0.48	0.017949	0.018100	1.51e-04	[0.017292, 0.018945]	[0.017283, 0.018945]	0.36
75	0.48	0.134902	0.134130	7.72e-04	[0.132032, 0.136256]	[0.132023, 0.136258]	-0.71
25	0.50	0.250000	0.250110	1.10e-04	[0.247435, 0.252804]	[0.247428, 0.252806]	0.08
50	0.50	0.500000	0.498460	1.54e-03	[0.495361, 0.501559]	[0.495356, 0.501564]	-0.97
75	0.50	0.750000	0.752640	2.64e-03	[0.749956, 0.755305]	[0.749953, 0.755312]	1.93
25	0.52	0.865098	0.866350	1.25e-03	[0.864227, 0.868445]	[0.864225, 0.868453]	1.16
50	0.52	0.982051	0.981580	4.71e-04	[0.980728, 0.982395]	[0.980727, 0.982404]	-1.12
75	0.52	0.997862	0.997830	3.25e-05	[0.997522, 0.998100]	[0.997522, 0.998109]	-0.22

Table 6 reports the comparison. The largest absolute discrepancy across the 9 cells is 2.64×10^{-3} , the largest standardised deviation is **1.93** standard errors, and the exact value falls inside the Wilson interval in 9 of 9 cells. There is no evidence of systematic bias: the standardised deviations are scattered on both sides of zero with a mean of **0.140**. The simulation and the algebra describe the same process.

9.3 Convergence and the inverse square root envelope

The standard error of a sample proportion is $\sqrt{P(1-P)/n}$, so the error of the Monte Carlo estimate should shrink like $n^{-1/2}$. Figure 4 tracks a single estimate as replications accumulate and overlays the theoretical envelope.

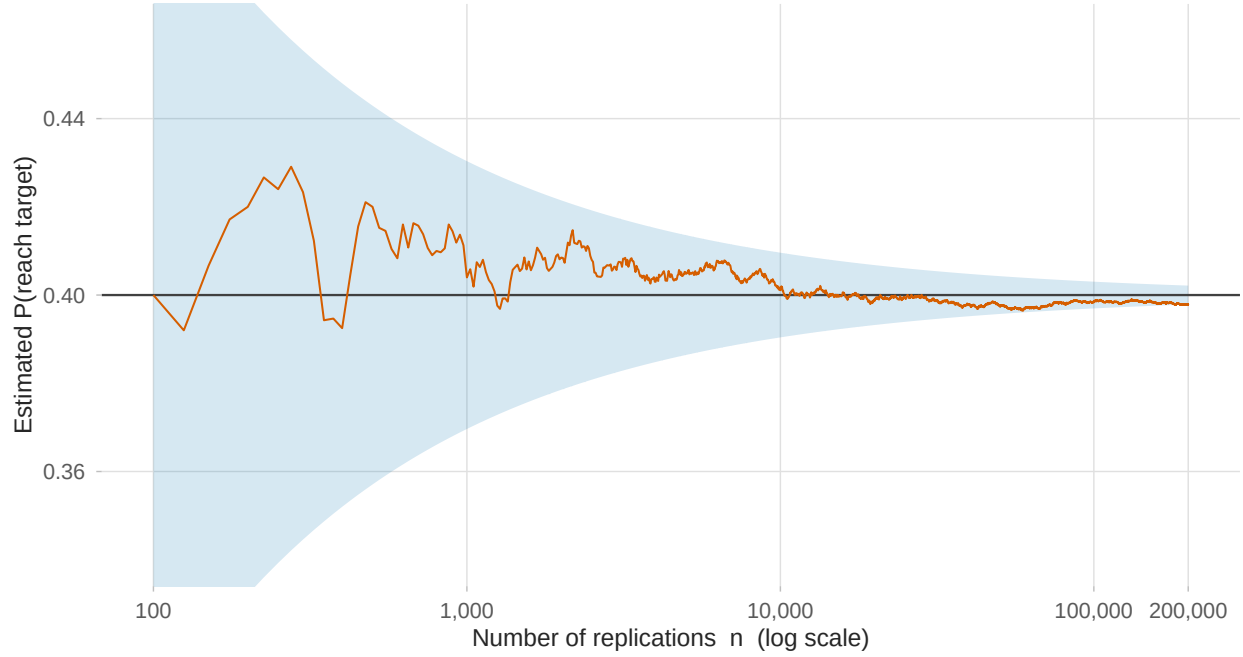


Figure 4: Running Monte Carlo estimate of the probability of reaching a target of 100 units from a starting capital of 40 in a fair game, whose exact value is 0.40. The shaded band is the theoretical 95 per cent envelope, which narrows in proportion to the inverse square root of the number of replications. The estimate settles inside the band and stays there, which is the behaviour a correct sampler should show.

After 200,000 replications the estimate differs from the exact value of **0.40** by 2.20×10^{-3} , and the running estimate lies inside the nominal 95 per cent envelope for 98.5% of the trajectory after the first hundred replications. The envelope narrows as a straight line would on a log-log plot, the signature of the $n^{-1/2}$ rate. The practical implication is the familiar one: reducing Monte Carlo error by a factor of ten costs a hundredfold increase in computation, which is why the closed forms remain valuable even when simulation is cheap.

9.4 Expected duration

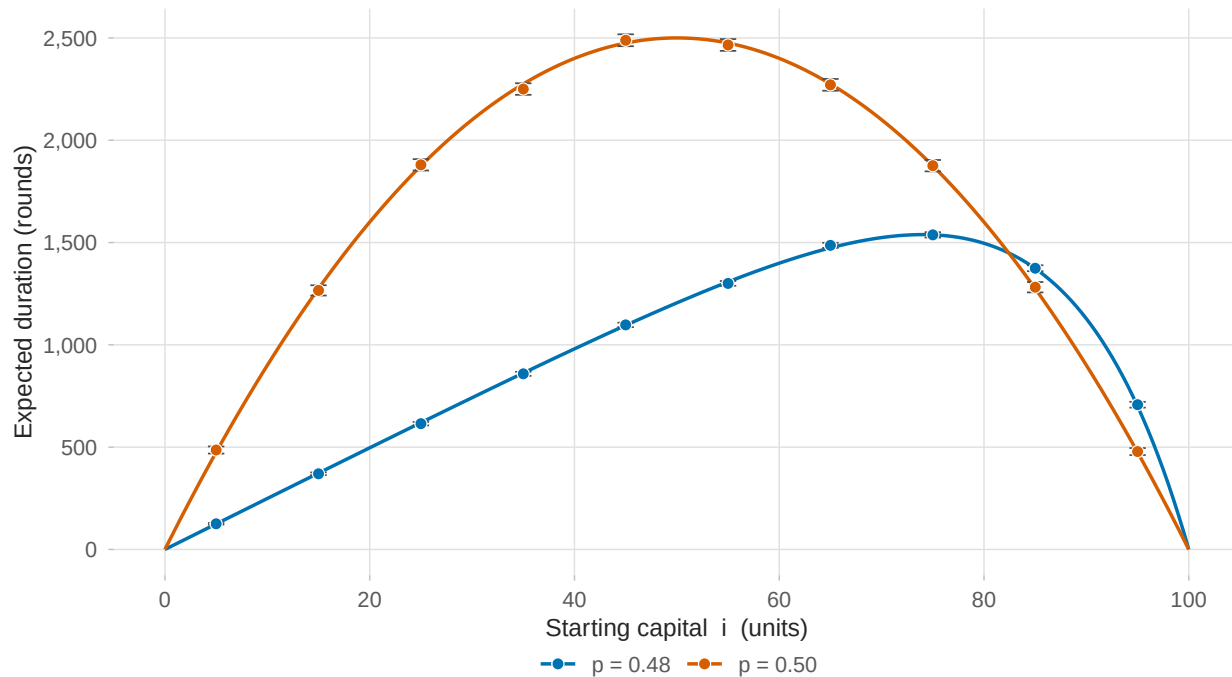


Figure 5: Expected duration of play against starting capital for a target of 100 units. Curves are the closed forms of Section 5; points are Monte Carlo means from 20,000 replications each, with vertical bars at plus and minus two standard errors and mostly smaller than the plotting symbols. The fair game is the symmetric parabola peaking at 2,500 rounds; a two percentage point disadvantage both lowers and skews it, because the drift carries the gambler to the lower barrier before the walk has time to explore.

Figure 5 overlays the simulated means on the analytical curves. The largest standardised deviation across the 20 cells is **1.97** standard errors. The fair-game curve is the parabola $i(N - i)$ of Proposition 4.2, symmetric about $i = 50$ where it attains **2,500** rounds. The disadvantaged curve is both lower and asymmetric, peaking at a starting capital of **74** units rather than at the midpoint, since a gambler who starts close to the target is carried away from it and a gambler who starts close to ruin reaches it quickly.

9.5 The distribution of durations

The expected duration is a poor summary of how long play actually lasts, because the distribution is severely right-skewed. Figure 6 shows the sampled durations for the fair game started at half the target.

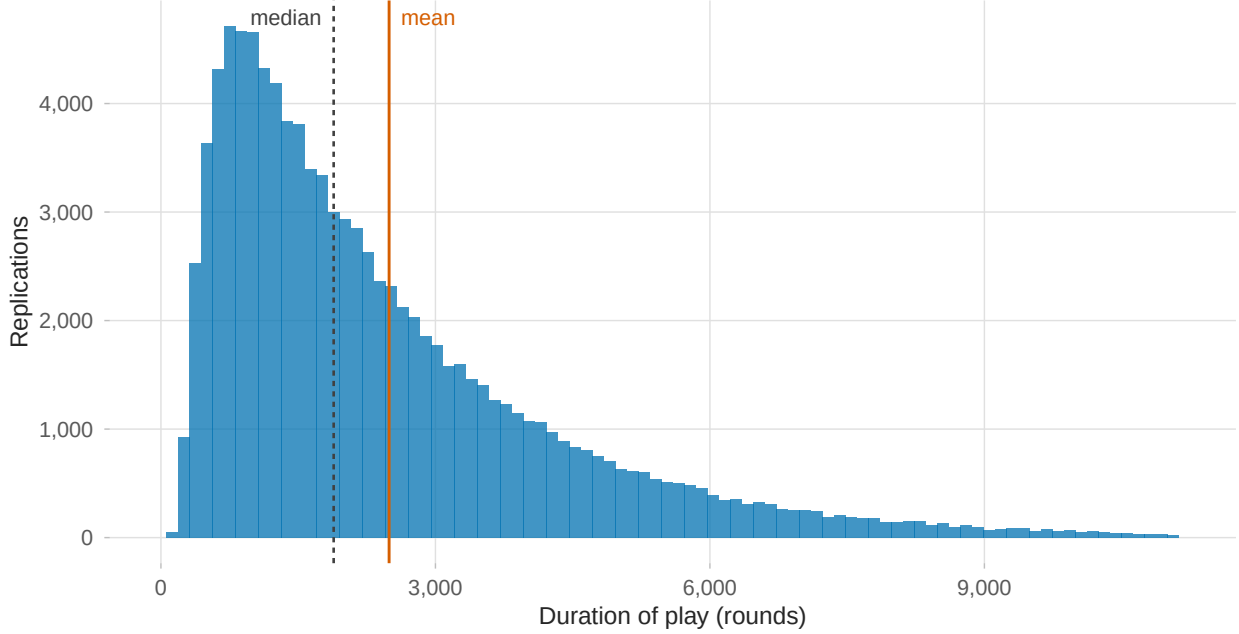


Figure 6: Distribution of the duration of play over 100,000 replications of a fair game started at 50 units with a target of 100. The distribution is extremely right-skewed: the median is well below the mean, and the longest simulated game ran many times longer still. A player budgeting time or risk capital by the average would be badly wrong about the tail.

The mean duration is **2,494** rounds against a median of only **1,888**, the ninetieth percentile is **5,156** and the ninety-ninth is **9,762**, while the longest of the 100,000 simulated games ran for **24,872** rounds. The exact mean from Proposition 4.2 is **2,500**, which the simulation matches to within 0.25%. The gap between mean and median is the reason that expected duration is an unreliable planning quantity, and it anticipates a theme of the empirical section: summary statistics computed under an idealised model can be technically correct and practically misleading at the same time.

10 Empirical application to equity index returns

The preceding sections established what the model says. This section asks what it is worth. The exercise has three parts: estimate the model’s single parameter from market data, test whether the independence assumption that justifies the model survives contact with that data, and then quantify how much the answer changes when the model’s most artificial assumption is relaxed.

10.1 Data

The data are daily closing levels of the S&P 500 composite index, retrieved programmatically and cached locally so that the report compiles without network access. The source used for this compilation was Yahoo Finance, S&P 500 composite index (^GSPC), daily close (Yahoo Finance 2026). The retrieval routine attempts Yahoo Finance first through the `quantmod` package (Ryan and Ulrich 2024) and falls back to the Federal Reserve Bank of St. Louis series SP500 (Federal Reserve Bank of St. Louis 2026) if that fails, with the source recorded in a provenance file alongside the cached data and reported here from that file rather than assumed. The sample runs from 02 January 1990 to 18 September 2026 and comprises 9,246 closing levels, giving 9,245 daily log returns $r_t = \log(C_t/C_{t-1})$, of which 5 are exactly zero and are excluded from the sign analysis.

Table 7: Summary statistics for daily S&P 500 log returns over the sample period. The excess kurtosis is the quantity of interest for what follows: a normal distribution would give zero, and the observed value indicates a distribution in which extreme days are far more common than any model with fixed step size can accommodate.

Statistic	Value
Observations	9,245
Mean daily log return	0.000331
Standard deviation	0.011334
Mean absolute return	0.007570
Skewness	-0.364
Excess kurtosis	10.877
Minimum	-0.1277
Maximum	0.1096

Table 7 records the basic shape of the data. The excess kurtosis of **10.9** against zero for a normal distribution, and the negative skewness of **-0.36**, are the two features that Section 10.4 shows to matter most.

10.2 Testing the random walk hypothesis on the sign sequence

The model of Section 2 assumes that the increments are independent and identically distributed. Applied to markets this is the random walk hypothesis in the form associated with Fama (1970). Four tests are reported.

The empirical fraction of up days is $\hat{p} = 0.5369$, with a Wilson 95 per cent interval of $[0.5267, 0.5471]$ and a Clopper-Pearson interval of $[0.5267, 0.5471]$. Both exclude one half comfortably, so over this sample the index rose on more days than it fell, by a margin that sampling variation does not explain.

The Wald-Wolfowitz runs test (Wald and Wolfowitz 1940) asks whether the up and down days are arranged in the pattern that independence would produce, by counting maximal blocks of consecutive identical signs. The sample contains **4,738** runs against an expected **4,596** under independence, giving $z = 2.974$ and $p = 0.0029$. The excess of runs, rather than a deficit, indicates that signs alternate slightly more often than chance would dictate, which is short-horizon mean reversion rather than momentum. The lag-one autocorrelation of the sign series is **-0.0311**, consistent in direction and small in magnitude.

The Ljung-Box test (Ljung and Box 1978) examines the first 10 autocorrelations jointly. On the sign series it gives $Q = 23.39$ on 10 degrees of freedom with $p = 0.0094$; on the returns themselves, $Q = 91.55$ with $p < 0.0001$. Both reject independence. For contrast, the same test on absolute returns gives $Q = 8,314$, which is larger by a factor of **355** and is the fingerprint of volatility clustering: the magnitudes of returns are strongly dependent even where the signs are only weakly so.

Table 8: Lo-MacKinlay variance ratio tests on daily log returns at aggregation horizons of 2 to 32 days. A ratio below one indicates mean reversion. The homoskedastic statistic z_1 and the heteroskedasticity-consistent statistic z_2 tell noticeably different stories, and the robust column is the one to believe for return data.

q	$VR(q)$	z_1 (homoskedastic)	p -value	z_2 (robust)	p -value
2	0.9213	-7.566	< 0.0001	-3.384	0.0007
4	0.8752	-6.415	< 0.0001	-2.784	0.0054
8	0.8039	-6.375	< 0.0001	-2.805	0.0050
16	0.7739	-4.939	< 0.0001	-2.229	0.0258
32	0.7404	-3.913	< 0.0001	-1.870	0.0615

The Lo-MacKinlay variance ratio test (Lo and MacKinlay 1988; Campbell et al. 1997) compares the variance of q -day returns with q times the variance of one-day returns; a random walk implies a ratio of one at every horizon. Table 8 reports ratios below one at every horizon tested, falling from **0.921** at 2 days to **0.740** at

32 days, which again indicates mean reversion. The implementation used here is written directly from the original paper rather than taken from a package, and was validated before use by simulation: on independent normal data both statistics reject at close to their nominal size, while on simulated data with volatility clustering the homoskedastic statistic over-rejects and the robust statistic holds its size, which is the behaviour the robust correction exists to provide.

Table 9: Calibration of the variance ratio implementation on 400 simulated samples of 2,000 observations each, none of which contains any predictability. A correctly sized test rejects 5 per cent of the time. Under volatility clustering the homoskedastic statistic rejects far too often while the robust statistic holds its size, which is the behaviour the robust correction exists to deliver and is the evidence that the implementation is correct.

Data generating process	Mean VR(4)	Rejection rate, z_1	Rejection rate, z_2
Independent normal	1.0007	5.8%	5.0%
Independent Student t(3)	0.9993	5.0%	5.0%
GARCH(1,1), no predictability	1.0027	9.2%	5.0%

Table 9 reports that calibration. On independent normal data both statistics reject at close to the nominal 5 per cent, confirming correct size. On simulated data with volatility clustering but no predictability whatsoever, the homoskedastic statistic rejects 9.2% of the time, a false positive rate more than 1.8 times its nominal level, while the robust statistic rejects 5.0% of the time. A further internal check is that the robust variance estimator reduces to the homoskedastic one when the data are in fact homoskedastic, which it does to within sampling error. This matters because secondary presentations of the Lo-MacKinlay statistic differ in where they place the sample size normalisation, and an implementation that is out by a factor of $\sqrt{T}$ would pass casual inspection while rejecting almost always. The calibration above is what distinguishes the two conventions.

That distinction matters for the interpretation. The homoskedastic statistics in Table 8 are very large, with $|z_1|$ reaching **7.6**, but a substantial part of that apparent significance is an artefact of changing variance rather than evidence of predictability. The robust statistics are far smaller, and at the longest horizon tested the robust test does not reject at the 5 per cent level ($z_2 = -1.87$, $p = 0.0615$) even though the homoskedastic test rejects overwhelmingly.

Table 10: Summary of tests of the random walk hypothesis. The statistical verdict is a rejection, but the magnitudes involved are small, and the strongest dependence in the data is in the size of returns rather than in their direction.

Test	Statistic	p-value	Conclusion
Proportion of up days ($\hat{p} = 0.5?$)	7.095	< 0.0001	Reject fairness
Wald-Wolfowitz runs test on signs	2.974	0.0029	Reject independence
Ljung-Box (10 lags) on signs	23.39	0.0094	Reject independence
Ljung-Box (10 lags) on returns	91.55	< 0.0001	Reject independence
Ljung-Box (10 lags) on absolute returns	8,314	< 0.0001	Strong dependence in magnitudes
Variance ratio, $q = 2$ (robust)	-3.384	0.0007	Reject random walk
Variance ratio, $q = 32$ (robust)	-1.870	0.0615	Does not reject at 5%

Table 10 collects the results, and the honest reading is mixed. Every test of the sign sequence rejects independence at conventional levels, so as a strict statistical hypothesis the random walk is dead. But the effect sizes are small. The runs test detects an excess of only 3.1% over the expected number of runs; the lag-one sign autocorrelation is **-0.0311**, which explains 0.096% of the variation in the next day's direction. With 9,240 observations, a departure this small is easily detectable while being far too weak to trade against after costs. The evidence is therefore that the sign sequence is not exactly a fair coin, and also that it is very nearly one. The stronger and less ambiguous finding is the dependence in magnitudes, which the plus or minus one model cannot represent at all because it fixes the magnitude by construction.

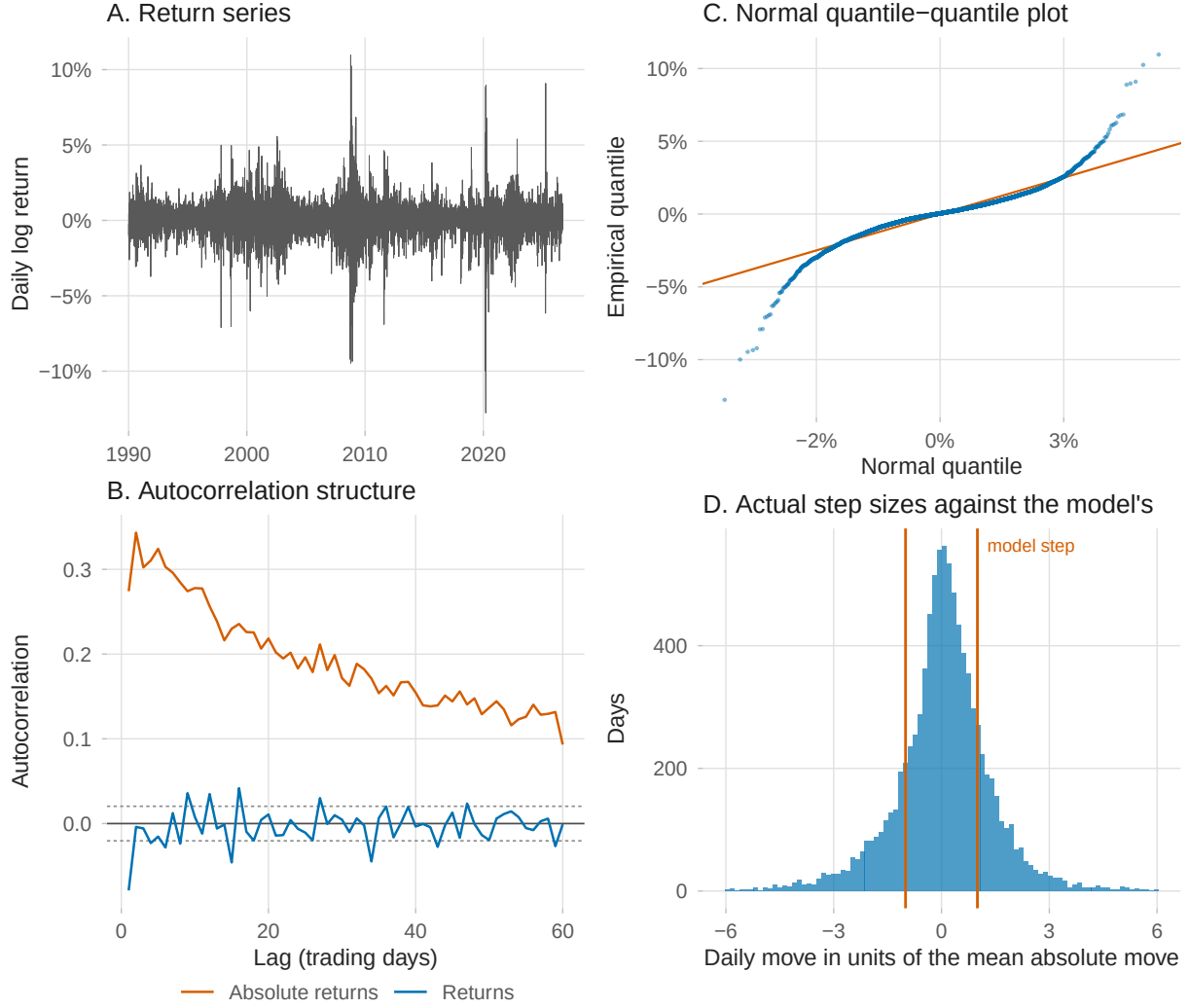


Figure 7: Diagnostics for the plus or minus one random walk model applied to daily S&P 500 returns. Panel A shows the return series, where the clustering of large moves into distinct episodes is visible directly. Panel B shows that return autocorrelations are negligible while absolute return autocorrelations decay slowly and remain far outside the confidence bands for months. Panel C compares the empirical quantiles with the normal, showing tails much heavier than Gaussian. Panel D contrasts the fixed unit step of the model with the actual distribution of standardised daily moves. Together the panels show that the model's failure is about the size of steps, not their direction.

10.3 Applying the ruin formulae

Consider a trader who stakes a fixed amount on the direction of the index each day, gaining one unit when the index rises and losing one when it falls, starting with i units of capital. This is exactly the model of Section 2 with $p = \hat{p}$. Because the trader faces a market of effectively unlimited depth and has no natural target at which to stop, Theorem 7.1 is the relevant result: with $p > 1/2$ the probability of eventual ruin is $(\hat{q}/\hat{p})^i$, and requiring this to fall below a threshold α gives

$$i \geq \frac{\log \alpha}{\log(\hat{q}/\hat{p})}. \quad (15)$$

Table 11: Capital required for the modelled probability of eventual ruin to fall below 5 and 1 per cent, against an opponent of unlimited capital, for win rates spanning the estimate $\hat{p} = 0.5369$. One unit is the amount staked per day. The final column shows how exposed a trader with 20 units would be at each win rate.

p	Edge $2p - 1$	Units for 5% ruin	Units for 1% ruin	Exact $\log \alpha / \log(q/p)$	Ruin with 20 units
0.5100	2.00%	75	116	115.11	44.93%
0.5200	4.00%	38	58	57.53	20.17%
0.5300	6.00%	25	39	38.33	9.05%
0.5369	7.38%	21	32	31.14	5.20%
0.5450	9.00%	17	26	25.52	2.71%
0.5600	12.00%	13	20	19.10	0.80%

Table 11 applies (15) across a range of win rates spanning the estimate. At the point estimate $\hat{p} = 0.5369$ the modelled requirement is **32** units of capital for the probability of eventual ruin to fall below 1 per cent. The sensitivity is severe. At the lower end of the Wilson interval, $p = 0.5267$, the requirement rises to **44** units, and at the upper end, $p = 0.5471$, it falls to **25** units. The entire span of the confidence interval on a single parameter changes the modelled capital requirement by a factor of roughly **1.8**. This is a general feature of (15) rather than an artefact of this sample: the requirement behaves like $\log(1/\alpha)/(4\varepsilon)$ for small edges $\varepsilon = p - 1/2$, so it diverges hyperbolically as the edge vanishes. Any capital requirement computed this way inherits the full uncertainty in $\hat{p}$ and should be quoted as a range.

These are modelled quantities under a deliberately simple description of trading, not forecasts. Whether they mean anything depends on whether the description is adequate, and the next section shows that it is not.

10.4 Relaxing the fixed step: a bootstrap comparison

The plus or minus one model makes three assumptions about the increments: that they have fixed magnitude, that they are independent, and that their distribution does not change over time. The tests above show the second and third to be imperfect. The first is not even approximately true, as panel D of Figure 7 makes plain. This section replaces the fixed step with the empirical distribution of returns and measures how much the answer moves.

The experiment is constructed so that the comparison is fair. Let $u = \frac{1}{T} \sum_t |r_t| = 0.00757$ be the mean absolute daily return, and define the standardised step $s_t = r_t/u$, so that the empirical steps have mean absolute size exactly one. Three specifications are then simulated through the identical engine and the identical barriers, and differ only in how increments are drawn.

The first is the idealised model, drawing $+1$ with probability $\hat{p}$ and -1 otherwise. The second is an independent bootstrap (Efron 1979), drawing s_t with replacement from the empirical sample, which preserves the fat tails and the asymmetry of the return distribution but destroys any serial dependence. The third is a moving-block bootstrap (Künsch 1989), drawing blocks of 10 consecutive standardised returns and laying them end to end, which retains volatility clustering within each block. All three share the same mean absolute step size of one unit and, by construction, the first two share the same probability of an up day.

Table 12: Projected probability of ruin for a trader starting with 20 units and a target of 100 units, where one unit is the mean absolute daily move. All three specifications use the same simulation engine, the same barriers and the same mean absolute step size; they differ only in the distribution of the daily step. The idealised row agrees with its closed form to within simulation error.

Specification	Modelled ruin probability	Wilson 95% interval	Mean duration (days)	Median duration (days)
Idealised ± 1 model	4.99%	[4.80%, 5.19%]	1,014	954
IID bootstrap	42.66%	[42.23%, 43.10%]	860	691
Block bootstrap	37.18%	[36.76%, 37.60%]	978	846

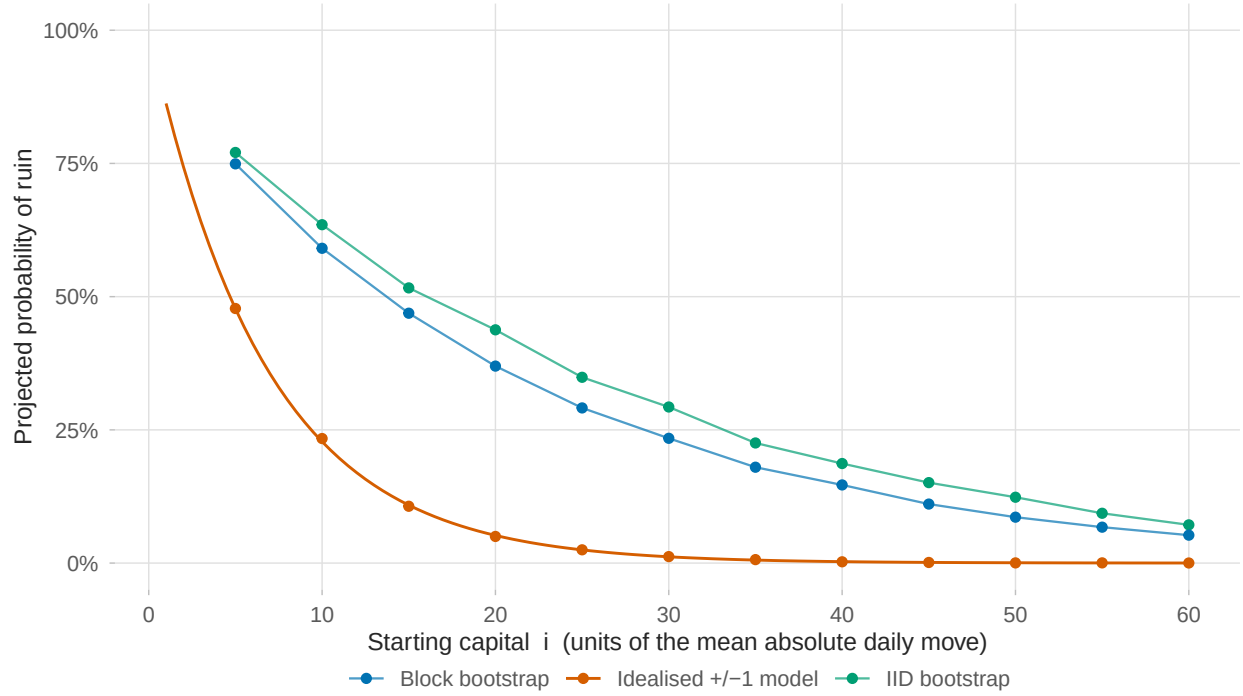


Figure 8: Projected probability of ruin against starting capital under the three specifications, with a target of 100 units. The idealised model, which knows only the frequency of up days, is far more optimistic than either bootstrap at every level of capital. The gap is the cost of assuming that every day moves the account by the same amount. Points are simulated from 12,000 replications per specification and starting capital; the line for the idealised model is its closed form.

The results in Table 12 are the central applied finding of this report. Starting from 20 units, the idealised model projects a ruin probability of **5.0%**, in agreement with the closed-form value of **5.2%** from Theorem 7.1 applied with a finite barrier. Replacing the fixed step with an independent bootstrap of actual returns raises that projection to **42.7%**, a factor of **8.5**. The block bootstrap, which additionally preserves volatility clustering, projects **37.2%**. Figure 8 shows that the gap persists across the whole range of starting capital.

Two mechanisms drive the difference, and they can be separated. The first is drift. The idealised model infers a per-round expected gain of $2\hat{p} - 1 = 0.0738$ units from the frequency of up days alone, whereas the actual standardised returns have mean 0.0437 units. The index rises on more days than it falls but falls by more when it does, so a model calibrated only to direction overstates the edge by a factor of **1.69**. The second is dispersion. The idealised step has standard deviation 0.997 by construction, while the standardised empirical step has standard deviation **1.497**, so the empirical walk diffuses roughly **1.50** times faster and reaches the lower barrier sooner. A smaller edge working against a larger variance is exactly the combination that Theorem 7.1 punishes.

Table 13: Sensitivity of the block bootstrap to the block length, at 20,000 replications per row. The projection is stable for blocks of 5 days or more and consistently below the independent bootstrap, so the direction of the effect is not an artefact of the particular block length chosen.

Block length L (days)	Projected ruin probability	Standard error
1 (iid)	42.54%	0.35%
5	38.22%	0.34%
10	37.04%	0.34%
20	37.13%	0.34%
40	37.73%	0.34%
60	37.73%	0.34%

The direction of the block bootstrap result deserves comment, since preserving dependence lowers rather than raises the projected ruin probability. Table 13 confirms that this is stable across block lengths from 5 to 60 days and is therefore not an artefact of the choice. The likely explanation is the mean reversion detected by the runs test and the variance ratio test: resampling in blocks preserves the tendency of moves to alternate, which dampens the accumulation of consecutive losses that drives a walk to the lower barrier. Independent resampling discards that structure and so permits longer adverse runs than the data actually exhibit. The effect is modest next to the eightfold increase caused by relaxing the fixed step size, and the honest summary is that step size distribution dominates serial dependence by a wide margin in this application.

11 Limitations and directions for extension

11.1 Limitations

The mathematics of Sections 2 to 9 is exact, and nothing in the empirical section casts doubt on it. What the empirical section casts doubt on is the mapping from that mathematics to a trading account, and the limitations are best stated in that order.

The fixed step size is the most consequential assumption and the least defensible. Section 10.4 quantified the damage: holding the win rate and the mean absolute move fixed and changing only the shape of the step distribution moved the projected ruin probability from **5.0%** to **42.7%**. Since real returns have excess kurtosis of **10.9** and negative skewness, and since ruin is a tail event, a model that fixes the magnitude of every move cannot be expected to price tail risk correctly. Any figure derived from the plus or minus one model should be treated as a lower bound on risk rather than an estimate of it.

Independence fails, though gently. The runs test, the Ljung-Box test and the variance ratio test all reject at conventional levels, yet the magnitudes are small enough that the sign sequence is very nearly a fair coin. The practical significance of this rejection is much smaller than its statistical significance, and the report has tried to present both rather than either alone.

Stationarity fails less gently. The parameter $\hat{p}$ was estimated over 9,245 days spanning 36 years, and treating it as a constant is a strong assumption across a period containing several distinct volatility regimes. The clustering visible in panel A of Figure 7 and the Ljung-Box statistic of **8,314** on absolute returns are evidence that the conditional distribution changes materially through time, which the models of Engle (1982) and Bollerslev (1986) were introduced to capture and which no constant-parameter random walk can represent.

Several narrower caveats belong on the record. The win rate $\hat{p}$ is estimated in sample and then used to make statements about the same sample, so the projections inherit no out-of-sample validation. The trading description ignores transaction costs, financing and slippage, each of which would lower the effective win rate, and Table 11 shows that the capital requirement is acutely sensitive to exactly that quantity. The bootstrap arms absorb whenever capital falls to zero or below, so a large adverse move can overshoot the barrier rather than land on it, which is realistic but means the barrier is not hit exactly as in the discrete model. The moving-block bootstrap restarts its blocks at the boundary of each simulation block, which breaks a small

fraction of blocks and mildly understates long-range dependence. Finally, the index series is not investable, carries no dividends, and is subject to whatever revisions and survivorship conventions the provider applies.

11.2 Directions for extension

Four extensions seem to me genuinely worth pursuing, in roughly increasing order of difficulty.

11.2.1 General increment distributions

The natural generalisation replaces the plus or minus one increment with an arbitrary distribution of finite mean, which is the setting of classical risk theory (Asmussen and Albrecher 2010). The analogue of the ratio q/p is the adjustment coefficient, or Lundberg exponent, defined as the positive root R of the equation

$$\mathbb{E}[e^{-RX}] = 1,$$

where X is the increment. When R exists, the probability of eventual ruin from capital x against an unlimited opponent satisfies the Lundberg inequality $\psi(x) \leq e^{-Rx}$, with the Cramer-Lundberg approximation $\psi(x) \sim Ce^{-Rx}$ under regularity conditions. The plus or minus one walk is the special case in which $e^{-R} = q/p$, so Theorem 7.1 is the elementary instance of a general theory. Crucially, R fails to exist when the increment distribution has a heavy enough left tail, and in that regime ruin probabilities decay polynomially rather than exponentially. Given the excess kurtosis measured in Section 10, determining whether an adjustment coefficient exists for standardised equity returns is the obvious next question, and it would put the bootstrap results of Section 10.4 on a theoretical footing rather than a purely numerical one.

11.2.2 Reflecting rather than absorbing barriers

Replacing absorption at zero with reflection changes the problem qualitatively. Adopt the convention that from state 0 the walk moves to 1 with probability p and remains at 0 with probability q , and symmetrically at N . The chain is then irreducible and aperiodic on a finite state space, hence ergodic, and the question shifts from absorption to the long-run distribution. Detailed balance for this birth-death chain requires $\pi_k p = \pi_{k+1} q$, giving the truncated geometric stationary distribution

$$\pi_k = \frac{(p/q)^k}{\sum_{j=0}^N (p/q)^j}, \quad k = 0, 1, \dots, N,$$

which reduces to the uniform distribution when $p = q$. The financial reading is a fund that is recapitalised at zero rather than shut down, and the relevant risk measure becomes the long-run fraction of time spent near the lower barrier rather than a probability of eventual ruin.

The stationary distribution computed from the dominant left eigenvector of the reflecting transition matrix agrees with the truncated geometric formula to 1.14×10^{-15} , which confirms the claim.

11.2.3 Three or more players

With $k \geq 3$ players the state is a vector of holdings constrained to a simplex, and the walk moves on a lattice polytope rather than an interval. No closed form comparable to Proposition 3.1 is known in general. The symmetric three-player case admits partial results, and symbolic and computational approaches have been used to generate exact answers for small configurations. The interesting structural question is which player is eliminated first, and the answer is not simply the one with least capital, because the order of elimination depends on the pairwise matching rules. This is the natural model for competing firms drawing on shared liquidity.

11.2.4 Variable stake sizes

The assumption that the stake is one unit regardless of capital is a modelling choice, not a law. Betting a fixed fraction of current wealth makes ruin in finite time impossible while introducing a different failure mode, namely asymptotic decay of wealth when the fraction is chosen badly, and the optimal fraction is the Kelly criterion (Kelly 1956). Under fractional betting the log of wealth rather than wealth itself performs an additive random walk, so the machinery of this report transfers directly after a change of variable, with ruin reinterpreted as falling below a threshold rather than reaching zero. Comparing the ruin and growth characteristics of fixed-stake and fixed-fraction schemes on the same empirical return distribution used in Section 10.4 would be a natural continuation of this work.

11.3 Data and software credits

The price data are daily closing levels of the S&P 500 composite index obtained from Yahoo Finance, S&P 500 composite index (^GSPC), daily close (Yahoo Finance 2026), retrieved on 2026-09-19 and cached in the accompanying `data/` directory so that the analysis reproduces without network access. The Federal Reserve Bank of St. Louis series SP500 (Federal Reserve Bank of St. Louis 2026) is configured as an automatic fallback source. The index itself is a product of S&P Dow Jones Indices; it is used here for non-commercial research and is not investable directly.

All computation was carried out in R (R Core Team 2026). The report was written in R Markdown and compiled with `bookdown` (Xie 2016). Figures use `ggplot2` (Wickham 2016) with the Okabe-Ito colour palette. Market data were retrieved with `quantmod` (Ryan and Ulrich 2024), and the runs test was cross-checked against `tseries` (Trapletti and Hornik 2024). The variance ratio test was implemented directly from Lo and MacKinlay (1988) rather than taken from a package, and the derivations in Sections 3 and 4 follow the classical treatment in Feller (1968), checked against Grimmett and Stirzaker (2001), Ross (2014) and Orosi (2018). Full version information is recorded in Appendix B.

12 Verification summary

Every analytical result in this report is compared against at least one independently computed value, and each comparison is recorded as a check with an explicit tolerance fixed before the comparison was made. Table 14 lists them all.

The checks are not decorative. Two of them failed during the preparation of this report and both failures were substantive. The martingale check failed because the probability of absorption at the origin was being formed as $1 - P_i$, which loses most of its significant digits when P_i is close to one; that is the episode described in Section 5, and it led to the stable evaluation used throughout. The variance ratio calibration failed because the heteroskedasticity-consistent statistic carried a spurious factor of $\sqrt{T}$, which made it reject on almost every independent normal sample; that is the error described in Section 10. Neither would have been visible from the output alone, since both produced numbers of entirely plausible magnitude.

A check that fails does not stop this document from being produced. Instead the quantity that depends on it is withheld: the text prints a marker in place of the number, and the table below carries the status. The reasoning is that a report which cannot be built communicates nothing, whereas a report that silently prints an unverified number communicates something false. Withholding the number is the only outcome that is both informative and honest.

Table 14: The 13 verification checks performed while compiling this report, with the tolerance each was held to and the value actually observed. Any failure is listed first, and the quantity depending on it is withheld from the text.

	Check	What it establishes	Observed	Tolerance	Status
12	bootstrap-baseline	Simulated idealised arm reproduces its own closed form	2.01e-03	1e-02	pass
7	convergence	Running estimate settles on the exact value	2.20e-03	1e-02	pass
3	diffusion	Discrete ruin probability converges to the Brownian hitting probability	8.57e-07	1e-03	pass
8	duration-grid	Simulated durations match the closed form across starting capitals	1.97e+00	4e+00	pass
1	martingale	Optional stopping identity agrees with the difference-equation solution	7.87e-15	1e-12	pass
2	matrix	Fundamental matrix reproduces both closed forms	1.77e-14	1e-10	pass
4	mc-coverage	Exact ruin probability inside the 99.9 per cent interval in every cell	0.00e+00	1e+00	pass
6	mc-duration	Simulated mean durations within 5 per cent of the closed form	3.72e-03	5e-02	pass
5	mc-z	Largest standardised Monte Carlo deviation within sampling range	1.93e+00	4e+00	pass
13	reflecting	Numerical stationary distribution matches the truncated geometric	1.14e-15	1e-10	pass
9	runs-test	Own runs-test implementation agrees with tseries::runs.test	0.00e+00	1e-08	pass
11	vr-robust	Robust rejection rate does not exceed the homoskedastic one under clustering	-4.25e-02	1e-02	pass
10	vr-size	Robust variance ratio statistic holds nominal size under independence	0.00e+00	3e-02	pass

All 13 checks passed in the compilation that produced this document, so no quantity has been withheld from the text above.

13 Conclusion

The gambler’s ruin problem admits a complete solution, and this report has given it three times over. The probability of reaching a target before ruin is $(1 - (q/p)^i)/(1 - (q/p)^N)$ in the asymmetric case and i/N in the symmetric case, the latter being the genuine limit of the former rather than a separate phenomenon; the expected duration is $i/(q - p) - NP_i/(q - p)$ and $i(N - i)$ respectively. Difference equations, optional stopping and numerical linear algebra agree to approximately fifteen decimal places, and a Monte Carlo study of 100,000 replications per cell agrees to within its own sampling error, with a largest standardised deviation of **1.93**. All 13 verification checks of Section 12 passed.

The structural conclusions are sharper than the formulae suggest. A gambler of finite means facing an unlimited opponent is ruined with probability one whenever the odds are fair or worse, and even a favourable game leaves a strictly positive ruin probability $(q/p)^i$ at every finite bankroll. The sensitivity of the ruin probability to a small edge is proportional to $i(N - i)/N$, which is to say proportional to the duration of the fair game, so the same edge that is invisible over a short session becomes decisive over a long one.

Applied to daily S&P 500 returns, the model estimates a win rate of $\hat{p} = 0.5369$ and finds that the sign sequence is statistically distinguishable from an independent one while being, in magnitude, very nearly independent. The more serious failure is elsewhere. When the fixed unit stake is replaced by a bootstrap of the actual return distribution, holding the win rate and the mean absolute move constant, the projected probability of ruin rises from **5.0%** to **42.7%**. The model’s weakness is not that markets are predictable but that they do not move in equal steps. That is a useful thing to know about a model, and it is the kind of thing that only becomes visible when the mathematics is exact enough that the discrepancy cannot be blamed on the algebra.

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A Appendix A: Full code listing

This appendix reproduces every line of code used to produce the report. The helper files are listed first, followed by the chunks in order of appearance. The listing is set in a smaller face, and the few lines too wide for the text block are broken at a space with a continuation indent; no code is omitted or altered. The authoritative copies are the files in the accompanying project directory.

A.1 Helper file: R/ruin.R

```
## -----
## ruin.R -- closed-form, matrix, and Monte Carlo machinery for the simple
## random walk with two absorbing barriers (the gambler's ruin problem).
##
## Notation follows the report throughout:
## i starting capital, 0 <= i <= N (units)
## N upper absorbing barrier (target capital, units)
## p probability of winning one unit on a single round
## q = 1 - p
## P_i probability of absorption at N before absorption at 0, from state i
## D_i expected number of rounds until absorption, from state i
## -----

## =====
## 1. Closed forms
## =====

#' Probability of reaching N before 0, starting from i.
#'
#' Evaluates  $P_i = (1 - (q/p)^i) / (1 - (q/p)^N)$  for  $p \neq 1/2$  and  $P_i = i/N$  for
#'  $p = 1/2$ . The implementation is written for numerical stability rather than
#' for literal resemblance to the algebra: the ratio is computed through
#' expm1 on the log scale, which avoids the catastrophic cancellation that
#' the naive expression suffers when  $q/p$  is close to unity, and switches to a
#' rescaled form when  $N * |\log(q/p)|$  is large enough to overflow.
#'
#' @param i Vector of starting capitals.
#' @param N Upper absorbing barrier (scalar).
#' @param p Win probability (scalar).
#' @param tol Half-width of the window around 1/2 treated as exactly fair.
#' @return Vector of probabilities, the same length as i.
ruin_prob <- function(i, N, p, tol = 1e-12) {
  stopifnot(length(N) == 1L, length(p) == 1L, N >= 1, p >= 0, p <= 1)
  i <- as.numeric(i)
  q <- 1 - p

  out <- if (abs(p - 0.5) < tol) {
    i / N
  } else if (p == 1) {
    as.numeric(i > 0)
  } else if (p == 0) {
    rep(0, length(i))
  } else {
    lr <- log(q) - log(p) # log(q/p)
    if (lr < 0 || N * lr < 500) {
      ## Stable whenever  $\exp(N * lr)$  does not overflow. Both expm1 calls carry
      ## full relative precision as  $lr \rightarrow 0$ , so the  $p \rightarrow 1/2$  limit is clean.
      expm1(i * lr) / expm1(N * lr)
    } else {
      ##  $q/p > 1$  with  $N * lr$  large: divide numerator and denominator by
      ##  $(q/p)^N$  before evaluating, so nothing overflows.
      a <- exp((i - N) * lr)
      b <- exp(-N * lr)
      (a - b) / (1 - b)
    }
  }
}
```

```

    }
  }

  out[i <= 0] <- 0
  out[i >= N] <- 1
  out
}

#' Probability of ruin (absorption at 0) starting from i.
#'
#' Mathematically this is just  $1 - P_i$ , but it is not computed that way. When
#'  $P_i$  is very close to one the subtraction  $1 - P_i$  destroys most of the
#' significant digits of the answer, because the information sits in the
#' digits that the subtraction cancels. The identity
#'
#'  $1 - P_i = (r^i - r^N) / (1 - r^N)$ ,  $r = q/p$ ,
#'
#' is evaluated directly instead, through expm1 on the log scale, which
#' keeps full relative precision no matter how close to one or zero the
#' answer is.
ruin_prob_0 <- function(i, N, p, tol = 1e-12) {
  stopifnot(length(N) == 1L, length(p) == 1L, N >= 1, p >= 0, p <= 1)
  i <- as.numeric(i)
  q <- 1 - p

  out <- if (abs(p - 0.5) < tol) {
    (N - i) / N
  } else if (p == 1) {
    as.numeric(i <= 0)
  } else if (p == 0) {
    rep(1, length(i))
  } else {
    lr <- log(q) - log(p)
    if (lr < 0 || N * lr < 500) {
      exp(i * lr) * expm1((N - i) * lr) / expm1(N * lr)
    } else {
      ## q/p > 1 with N * lr large: rescale by (q/p)^N before evaluating.
      (exp((i - N) * lr) - 1) / (exp(-N * lr) - 1)
    }
  }

  out[i <= 0] <- 1
  out[i >= N] <- 0
  out
}

#' Expected number of rounds until absorption, starting from i.
#'
#' Symmetric case  $D_i = i(N - i)$ ; asymmetric case
#'  $D_i = i/(q - p) - (N/(q - p)) * (1 - (q/p)^i)/(1 - (q/p)^N)$ , which is
#' evaluated in the algebraically identical but numerically kinder form
#'  $D_i = (i - N * P_i) / (q - p)$ , reusing ruin_prob.
ruin_duration <- function(i, N, p, tol = 1e-10) {
  stopifnot(length(N) == 1L, length(p) == 1L)
  i <- as.numeric(i)
  q <- 1 - p

  out <- if (abs(p - 0.5) < tol) {
    i * (N - i)
  } else {
    (i - N * ruin_prob(i, N, p)) / (q - p)
  }

  out[i <= 0 | i >= N] <- 0
  out
}

```

```

# ' Ruin probability against an infinitely rich opponent (N -> infinity).
# '
# ' Returns 1 when p <= 1/2 and (q/p)^i when p > 1/2.
ruin_prob_infinite <- function(i, p) {
  if (p <= 0.5) rep(1, length(i)) else ((1 - p) / p)^i
}

## =====
## 2. Markov chain in matrix form
## =====

# ' Full (N+1) x (N+1) transition matrix on states {0, 1, ..., N}.
# '
# ' States 0 and N are absorbing. Rows and columns are labelled by capital.
transition_matrix <- function(N, p) {
  stopifnot(N >= 2)
  q <- 1 - p
  P <- matrix(0, nrow = N + 1, ncol = N + 1,
              dimnames = list(as.character(0:N), as.character(0:N)))
  P[1, 1] <- 1 # state 0 absorbing
  P[N + 1, N + 1] <- 1 # state N absorbing
  for (k in 1:(N - 1)) { # state k sits in row k+1
    P[k + 1, k + 2] <- p
    P[k + 1, k] <- q
  }
  P
}

# ' Canonical decomposition of the chain into transient and absorbing blocks.
# '
# ' Returns Q (transient-to-transient, (N-1) x (N-1)) and R (transient-to-
# ' absorbing, (N-1) x 2, columns ordered as state 0 then state N).
canonical_parts <- function(N, p) {
  P <- transition_matrix(N, p)
  trans <- as.character(1:(N - 1))
  absb <- c("0", as.character(N))
  list(Q = P[trans, trans, drop = FALSE],
       R = P[trans, absb, drop = FALSE],
       P = P)
}

# ' Fundamental matrix M = (I - Q)^{-1} of the absorbing chain.
# '
# ' Entry M[i, j] is the expected number of visits to transient state j before
# ' absorption, starting from transient state i.
fundamental_matrix <- function(N, p) {
  cp <- canonical_parts(N, p)
  Q <- cp$Q
  solve(diag(nrow(Q)) - Q)
}

# ' Absorption probabilities and expected absorption times by linear algebra.
# '
# ' A purely numerical route to the same quantities as `ruin_prob` and
# ' `ruin_duration`, used in the report as an independent cross-check that
# ' shares no code path with the closed forms.
matrix_absorption <- function(N, p) {
  cp <- canonical_parts(N, p)
  M <- solve(diag(nrow(cp$Q)) - cp$Q)
  B <- M %*% cp$R # absorption probabilities
  t <- as.numeric(M %*% rep(1, nrow(M)))
  list(fundamental = M,
       prob_0 = stats::setNames(B[, 1], rownames(B)),
       prob_N = stats::setNames(B[, 2], rownames(B)),
       duration = stats::setNames(t, rownames(B)),
       states = 1:(N - 1))
}

```

```

}

## =====
## 3. Simulation
## =====

#' Simulate one random walk path to absorption.
#'
#' The path is generated in geometrically growing blocks and accumulated with
#' `cumsum`, so no scalar loop over rounds is ever executed; the outer loop
#' runs once per block, not once per step.
#'
#' @return A list with the full `path` (including the starting value),
#' the `absorbed` state (0 or N, or NA if `max_steps` was exhausted),
#' the `duration` in rounds, and a `hit_N` flag.
simulate_walk <- function(i, N, p, max_steps = 1e6, block = 256L) {
  pos <- i
  path <- i
  n <- 0L
  b <- block

  while (pos > 0 && pos < N && n < max_steps) {
    steps <- 2L * (stats::runif(b) < p) - 1L
    seg <- pos + cumsum(steps)
    hit <- which(seg <= 0 | seg >= N)
    if (length(hit)) seg <- seg[seq_len(hit[1L])]
    path <- c(path, seg)
    n <- n + length(seg)
    pos <- seg[length(seg)]
    b <- min(b * 2L, 65536L)
  }

  list(path = path,
       absorbed = if (pos <= 0) 0L else if (pos >= N) as.integer(N) else
         NA_integer_,
       hit_N = isTRUE(pos >= N),
       duration = length(path) - 1L)
}

#' Generic absorbed-walk simulation engine.
#'
#' Simulates `n` independent walks from `start`, absorbing at or beyond
#' `lower` and `upper`, where the increments are supplied by `step_sampler`,
#' a function of (rows, cols) returning a matrix of increments. Paths are
#' advanced a block of rounds at a time for every still-active walk
#' simultaneously, using `matrixStats::rowCumsums`; walks drop out of the
#' active set as they are absorbed, and the block length doubles so that the
#' heavy right tail of the duration distribution costs few iterations.
#'
#' @return data.frame with `hit_upper` (logical), `duration`, and `final`.
simulate_absorbed <- function(n, start, upper, step_sampler,
                             lower = 0, chunk = 5000L, block = 512L,
                             max_steps = 5e6) {
  hit_upper <- logical(n)
  duration <- numeric(n)
  final <- numeric(n)

  for (s in seq(1L, n, by = chunk)) {
    idx <- s:min(s + chunk - 1L, n)
    m0 <- length(idx)

    pos <- rep(start, m0)
    d <- numeric(m0)
    hitU <- logical(m0)
    active <- seq_len(m0)
    b <- block
  }
}

```

```

tot    <- 0

while (length(active) > 0L && tot < max_steps) {
  m      <- length(active)
  steps  <- step_sampler(m, b)
  paths  <- matrixStats::rowCumsums(steps) + pos[active]

  absorbed <- (paths <= lower) | (paths >= upper)
  any_abs  <- rowSums(absorbed) > 0

  if (any(any_abs)) {
    rows    <- which(any_abs)
    fc      <- max.col(absorbed[rows, , drop = FALSE], ties.method = "first")
    endpos  <- paths[cbind(rows, fc)]
    d[active[rows]] <- d[active[rows]] + fc
    hitU[active[rows]] <- endpos >= upper
    pos[active[rows]] <- endpos
  }

  keep <- which(!any_abs)
  if (length(keep)) {
    d[active[keep]] <- d[active[keep]] + b
    pos[active[keep]] <- paths[cbind(keep, rep(b, length(keep)))]
  }

  active <- active[keep]
  tot    <- tot + b
  b      <- min(b * 2L, 16384L)
}

hit_upper[idx] <- hitU
duration[idx]  <- d
final[idx]    <- pos
}

data.frame(hit_upper = hit_upper, duration = duration, final = final)
}

## Bernoulli (+/-1) step sampler.
bernoulli_sampler <- function(p) {
  function(m, b) matrix(2 * (stats::runif(m * b) < p) - 1, nrow = m, ncol = b)
}

## Monte Carlo study of the gambler's ruin for the +/-1 walk.
##
## @return data.frame of `hit_upper` (reached N) and `duration`.
simulate_ruin <- function(n, i, N, p, ...) {
  simulate_absorbed(n = n, start = i, upper = N,
    step_sampler = bernoulli_sampler(p), ...)
}

## =====
## 4. Bootstrap step samplers for the empirical section
## =====

## IID bootstrap sampler drawing with replacement from an empirical vector.
iid_boot_sampler <- function(x) {
  function(m, b) matrix(sample(x, m * b, replace = TRUE), nrow = m, ncol = b)
}

## Moving-block bootstrap sampler, preserving short-range dependence.
##
## Blocks of `L` consecutive observations are drawn with replacement and laid
## end to end, so volatility clustering inside a block survives resampling.
## Blocks restart at the boundary of each simulation block; with L much
## smaller than the simulation block length this affects a small fraction of

```

```

# rounds and is noted in the report's limitations.
block_boot_sampler <- function(x, L) {
  Tn <- length(x)
  stopifnot(L >= 1, L <= Tn)
  function(m, b) {
    nb <- ceiling(b / L)
    start <- matrix(sample.int(Tn - L + 1L, m * nb, replace = TRUE),
                     nrow = m, ncol = nb)
    idx <- matrix(0L, nrow = m, ncol = nb * L)
    for (l in seq_len(L)) idx[, seq(1, nb * L, by = L)] <- start + (l - 1L)
    matrix(x[idx], nrow = m)[, seq_len(b), drop = FALSE]
  }
}

## =====
## 5. Interval estimates
## =====

# Wilson score interval for a binomial proportion.
wilson_ci <- function(x, n, conf = 0.95) {
  z <- stats::qnorm(1 - (1 - conf) / 2)
  ph <- x / n
  den <- 1 + z^2 / n
  ctr <- (ph + z^2 / (2 * n)) / den
  hw <- z * sqrt(ph * (1 - ph) / n + z^2 / (4 * n^2)) / den
  data.frame(estimate = ph, lower = pmax(0, ctr - hw), upper = pmin(1, ctr +
    hw))
}

# Clopper-Pearson (exact) interval for a binomial proportion.
clopper_pearson_ci <- function(x, n, conf = 0.95) {
  a <- (1 - conf) / 2
  lo <- ifelse(x == 0, 0, stats::qbeta(a, x, n - x + 1))
  hi <- ifelse(x == n, 1, stats::qbeta(1 - a, x + 1, n - x))
  data.frame(estimate = x / n, lower = lo, upper = hi)
}

# Smallest starting capital with modelled ruin probability below `target`.
#
# Searches i = 1, 2, ... for the first capital at which the probability of
# ruin before reaching the barrier `N` falls below `target`. Returns NA if no
# capital below `max_i` achieves it.
capital_for_ruin_target <- function(p, target = 0.01, N = Inf, max_i = 5000L) {
  i <- seq_len(max_i)
  r <- if (is.infinite(N)) ruin_prob_infinite(i, p) else ruin_prob_0(i, N, p)
  hit <- which(r < target)
  if (!length(hit)) NA_integer_ else as.integer(i[hit[1L]])
}

## =====
## 6. Historical check
## =====

# Huygens' fifth problem (1657), solved from the general ruin formula.
#
# Players A and B hold `counters` counters each and throw three dice. A gains
# a counter on a throw of eleven, B on a throw of fourteen, and all other
# throws are disregarded; play continues until one player holds every
# counter. Conditioning on a decisive throw reduces this to the standard
# problem with i = counters and N = 2 * counters. The dice counts are
# enumerated rather than quoted.
huygens_odds <- function(counters = 12L) {
  sums <- as.vector(outer(outer(1:6, 1:6, "+"), 1:6, "+"))
  a <- sum(sums == 11L) # ways to throw eleven
  b <- sum(sums == 14L) # ways to throw fourteen

```

```

gcd <- function(x, y) if (y == 0) x else gcd(y, x %% y)
g   <- gcd(a, b)
p   <- a / (a + b)
list(ways_11 = a, ways_14 = b,
      p = p,
      prob_A = ruin_prob(counters, 2L * counters, p),
      base = c(a / g, b / g),
      num = (a / g)^counters,
      den = (b / g)^counters)
}

```

A.2 Helper file: R/tests.R

```

## -----
## tests.R -- tests of the random walk hypothesis used in the empirical section.
##
## The variance ratio statistic is implemented directly from Lo and MacKinlay
## (1988) rather than taken from a package, so that both the homoskedastic and
## the heteroskedasticity-consistent standard errors are available and their
## construction is visible.
## -----

#' Lo-MacKinlay variance ratio test.
#'
#' Given a series of one-period log returns, tests the null that the
#' underlying log price follows a random walk by comparing 1/q times the
#' variance of q-period returns with the variance of one-period returns. Under
#' the null the ratio is one.
#'
#' Two standard errors are reported. The statistic `z1` assumes homoskedastic
#' increments; `z2` uses the heteroskedasticity-consistent variance estimator,
#' which is the relevant one for financial returns because it remains valid
#' under volatility clustering, so that a rejection cannot be blamed on
#' changing variance alone.
#'
#' @param r Numeric vector of one-period log returns.
#' @param q Aggregation horizon (integer >= 2).
#' @return A one-row data.frame: q, VR, z1, p1, z2, p2.
lo_mackinlay_vr <- function(r, q) {
  stopifnot(q >= 2, q == as.integer(q))
  r <- as.numeric(r[!is.na(r)])
  nq <- length(r)                # number of one-period returns
  stopifnot(nq > q + 1)

  mu <- mean(r)
  e <- r - mu                    # demeaned one-period increments

  ## Variance of one-period returns (unbiased).
  sig_a <- sum(e^2) / (nq - 1)

  ## Variance of overlapping q-period returns, with the Lo-MacKinlay
  ## unbiasing constant m = q(nq - q + 1)(1 - q/nq).
  agg <- stats::filter(r, rep(1, q), sides = 1)
  agg <- as.numeric(agg[!is.na(agg)])      # nq - q + 1 overlapping sums
  m <- q * (nq - q + 1) * (1 - q / nq)
  sig_c <- sum((agg - q * mu)^2) / m

  VR <- sig_c / sig_a

  ## Homoskedastic standard error.
  v1 <- 2 * (2 * q - 1) * (q - 1) / (3 * q)
  z1 <- sqrt(nq) * (VR - 1) / sqrt(v1)

  ## Heteroskedasticity-consistent standard error:
  ## theta = sum_{j=1}^{q-1} [2(q-j)/q]^2 * delta_j

```

```

##   delta_j = sum_k e_k^2 e_{k-j}^2 / ( sum_k e_k^2 )^2
denom <- sum(e^2)^2
delta <- vapply(seq_len(q - 1), function(j) {
  sum(e[(j + 1):nq]^2 * e[1:(nq - j)]^2) / denom
}, numeric(1))
## theta already carries the sample-size normalisation, so unlike z1 the
## robust statistic takes no explicit sqrt(nq) factor.
theta <- sum((2 * (q - seq_len(q - 1)) / q)^2 * delta)
z2 <- (VR - 1) / sqrt(theta)

data.frame(q = q,
           VR = VR,
           z1 = z1, p1 = 2 * stats::pnorm(-abs(z1)),
           z2 = z2, p2 = 2 * stats::pnorm(-abs(z2)))
}

#' Wald-Wolfowitz runs test computed from first principles.
#'
#' Reported alongside `tseries::runs.test` so that the observed and expected
#' run counts, which carry the effect size, are available to the test rather
#' than only the test statistic.
#'
#' @param s Vector of signs (or any two-valued vector), zeros removed.
runs_summary <- function(s) {
  s <- s[s != 0]
  n1 <- sum(s > 0); n2 <- sum(s < 0); n <- n1 + n2
  runs <- 1 + sum(s[-1] != s[-n])
  mu <- 2 * n1 * n2 / n + 1
  v <- 2 * n1 * n2 * (2 * n1 * n2 - n) / (n^2 * (n - 1))
  z <- (runs - mu) / sqrt(v)
  data.frame(n = n, n_up = n1, n_down = n2,
            runs_observed = runs, runs_expected = mu, sd = sqrt(v),
            z = z, p_value = 2 * stats::pnorm(-abs(z)))
}

```

A.3 Helper file: R/data.R

```

## -----
## data.R -- reproducible retrieval and caching of daily equity index data.
##
## The report must build offline. Every download writes a CSV into data/ along
## with a small provenance record; subsequent builds read the cache and never
## touch the network. Deleting the cache forces a fresh download.
## -----

#' Retrieve daily S&P 500 closing levels, with an on-disk cache.
#'
#' Primary source is Yahoo Finance (^GSPC) via quantmod. If that fails for any
#' reason the function falls back to the Federal Reserve Bank of St. Louis
#' (FRED series SP500). The provenance record notes which source was used so
#' that the report test can report it rather than assert it.
#'
#' @param cache_dir Directory holding the cached CSV and provenance file.
#' @param from,to Date range requested (character or Date).
#' @param refresh If TRUE, ignore any existing cache and re-download.
#' @return A list with `data` (data.frame of date and close), `source`,
#' `downloaded_on`, and `note`.
load_sp500 <- function(cache_dir = "data",
                      from = "1990-01-01",
                      to = Sys.Date(),
                      refresh = FALSE) {

  csv_path <- file.path(cache_dir, "sp500_daily.csv")
  meta_path <- file.path(cache_dir, "sp500_provenance.dcf")

```

```

if (!refresh && file.exists(csv_path) && file.exists(meta_path)) {
  dat <- utils::read.csv(csv_path, stringsAsFactors = FALSE)
  dat$date <- as.Date(dat$date)
  meta <- as.list(read.dcf(meta_path)[1, ])
  return(list(data = dat,
             source = meta$Source,
             downloaded_on = meta$Downloaded,
             note = meta$Note,
             cached = TRUE))
}

if (!dir.exists(cache_dir)) dir.create(cache_dir, recursive = TRUE)

src <- NA_character_
note <- NA_character_
dat <- NULL

## --- Primary: Yahoo Finance -----
attempt <- try({
  px <- quantmod::getSymbols("^GSPC", src = "yahoo", from = from, to = to,
                             auto.assign = FALSE, warnings = FALSE)
  data.frame(date = as.Date(zoo::index(px)),
             close = as.numeric(quantmod::Cl(px)),
             stringsAsFactors = FALSE)
}, silent = TRUE)

if (!inherits(attempt, "try-error") && nrow(attempt) > 1000) {
  dat <- attempt
  src <- "Yahoo Finance, S&P 500 composite index (^GSPC), daily close"
  note <- "Retrieved with quantmod::getSymbols(src = 'yahoo')."
} else {
  ## --- Fallback: FRED -----
  attempt2 <- try({
    px <- quantmod::getSymbols("SP500", src = "FRED", auto.assign = FALSE)
    px <- px[!is.na(px)]
    px <- px[zoo::index(px) >= as.Date(from) & zoo::index(px) <= as.Date(to)]
    data.frame(date = as.Date(zoo::index(px)),
               close = as.numeric(px),
               stringsAsFactors = FALSE)
  }, silent = TRUE)
  if (inherits(attempt2, "try-error") || nrow(attempt2) < 100) {
    stop("Both the Yahoo Finance and FRED downloads failed, and no cached ",
         "copy of the price series is present in '", cache_dir, "'.")
  }
  dat <- attempt2
  src <- "FRED (Federal Reserve Bank of St. Louis), series SP500, daily close"
  note <- paste("Yahoo Finance was unavailable, so the FRED fallback was used.",
               "Note that FRED's SP500 series carries only a rolling",
               "ten-year",
               "window, so the sample begins later than the requested start",
               "date.")
}

dat <- dat[!is.na(dat$close) & dat$close > 0, ]
dat <- dat[order(dat$date), ]
rownames(dat) <- NULL

utils::write.csv(dat, csv_path, row.names = FALSE)
write.dcf(data.frame(Source = src,
                     Downloaded = as.character(Sys.Date()),
                     Rows = nrow(dat),
                     First = as.character(min(dat$date)),
                     Last = as.character(max(dat$date)),
                     Note = note,
                     stringsAsFactors = FALSE),

```

```

    meta_path)

  list(data = dat, source = src, downloaded_on = as.character(Sys.Date()),
        note = note, cached = FALSE)
}

```

A.4 Helper file: R/theme.R

```

## -----
## theme.R -- one plotting theme and one palette, defined once and reused by
## every figure in the report, plus small formatting helpers so that numbers
## quoted in the prose are produced by inline R rather than typed by hand.
## -----

suppressPackageStartupMessages({
  library(ggplot2)
  library(grid)
})

## Okabe-Ito qualitative palette: distinguishable under the common forms of
## colour vision deficiency and legible when printed in greyscale.
okabe_ito <- c("#0072B2", "#D55E00", "#009E73", "#CC79A7",
               "#E69F00", "#56B4E9", "#F0E442", "#000000")

theme_gr <- function(base_size = 9) {
  theme_minimal(base_size = base_size) +
    theme(
      panel.grid.minor = element_blank(),
      panel.grid.major = element_line(linewidth = 0.25, colour = "grey88"),
      panel.background = element_blank(),
      plot.background = element_blank(),
      axis.title       = element_text(size = base_size, colour = "grey15"),
      axis.text        = element_text(size = base_size - 1, colour =
        "grey35"),
      axis.ticks       = element_line(linewidth = 0.25, colour = "grey75"),
      axis.ticks.length = unit(2, "pt"),
      strip.text       = element_text(size = base_size, colour = "grey15",
        margin = margin(b = 3, t = 1)),
      strip.background = element_blank(),
      legend.position  = "bottom",
      legend.title     = element_text(size = base_size - 1, colour =
        "grey15"),
      legend.text      = element_text(size = base_size - 1, colour =
        "grey35"),
      legend.key.height = unit(9, "pt"),
      legend.key.width  = unit(16, "pt"),
      legend.margin    = margin(t = 0, b = 0),
      legend.box.spacing = unit(4, "pt"),
      plot.title       = element_text(size = base_size + 1, colour = "grey10",
        margin = margin(b = 4)),
      plot.margin      = margin(3, 6, 3, 3)
    )
}

scale_colour_gr <- function(...) ggplot2::scale_colour_manual(values =
  okabe_ito, ...)
scale_color_gr  <- scale_colour_gr
scale_fill_gr   <- function(...) ggplot2::scale_fill_manual(values = okabe_ito,
  ...)

## Colours reserved for the two absorbing outcomes, used consistently wherever
## outcome is encoded: reaching the target and being ruined.
col_target <- okabe_ito[1] # blue
col_ruin   <- okabe_ito[2] # vermillion
col_neutral <- "grey35"

```

```

set_gr_defaults <- function(base_size = 9) {
  ggplot2::theme_set(theme_gr(base_size))
  ggplot2::update_geom_defaults("line", list(linewidth = 0.45))
  ggplot2::update_geom_defaults("point", list(size = 1.1))
  invisible(NULL)
}

```

A.5 Helper file: R/utils.R

```

## -----
## utils.R -- formatting helpers.
##
## Every number that appears in the body text is produced by one of these
## functions applied to a computed object, so that the prose cannot drift out
## of step with the computation.
## -----

#' Fixed-decimal formatting with thousands separators.
fmt <- function(x, d = 3) formatC(x, format = "f", digits = d, big.mark = ",")

#' Integer formatting with thousands separators.
fmti <- function(x) formatC(round(x), format = "d", big.mark = ",")

#' Percentage formatting.
pct <- function(x, d = 1) paste0(formatC(100 * x, format = "f", digits = d),
  "\\%")

#' Significant-figure formatting.
sig <- function(x, d = 3) formatC(signif(x, d), format = "g", big.mark = ",")

#' Scientific notation rendered as LaTeX maths, e.g. 4.20 \times 10^{-7}.
sci <- function(x, d = 2) {
  if (!is.finite(x) || x == 0) return(fmt(x, d))
  e <- floor(log10(abs(x)))
  m <- x / 10^e
  paste0("$", formatC(m, format = "f", digits = d), " \\times 10^{", e, "}$")
}

#' p-values: small ones reported below a threshold rather than as zero.
fmt_p <- function(p, d = 4) {
  ifelse(p < 10^(-d), paste0("< ", formatC(10^(-d), format = "f", digits = d)),
    formatC(p, format = "f", digits = d))
}

#' Percentage formatting that refuses to print a bare 100 per cent.
#
#' For a finite barrier the ruin probability is strictly less than one, so a
#' value that merely rounds up to 100 per cent should not be displayed as
#' certainty. Such values are shown as a strict lower bound instead. Some of
#' them are indistinguishable from one in double precision, which is itself a
#' reason not to print them as an equality.
pct_strict <- function(x, d = 2) {
  thr <- 1 - 0.5 * 10^(-d - 2)
  lab <- formatC(100 * (1 - 10^(-d - 2)), format = "f", digits = d)
  ifelse(x >= thr, paste0("$>$ ", lab, "\\%"), pct(x, d))
}

#' p-value for inline prose, carrying its own relational operator.
#
#' Writing "p = " in the text and letting the formatter supply "< 0.0001"
#' produces "p < 0.0001". This returns the operator with the number so the
#' prose reads correctly whichever branch is taken.
fmt_p_rel <- function(p, d = 4) {
  ifelse(p < 10^(-d),

```

```

    paste0("< ", formatC(10^(-d), format = "f", digits = d)),
    paste0("= ", formatC(p, format = "f", digits = d)))
}

#' A confidence interval written as [lower, upper].
fmt_ci <- function(lo, hi, d = 4) paste0("[", fmt(lo, d), ", ", fmt(hi, d), "]")

#' Hard-wrap source lines for the printed code listing.
#'
#' The appendix reproduces the code verbatim, and a few lines (mostly long
#' table captions) are wider than the text block. Rather than let them run
#' into the margin, they are broken at the last space that fits, with a
#' continuation indent. Nothing is removed; only line breaks are inserted.
wrap_src_lines <- function(lines, width = 80L, cont = " ") {
  out <- character(0)
  for (ln in lines) {
    base <- sub("^(\\t)*.*$", "\\1", ln)
    cind <- paste0(base, cont)
    cur <- ln
    prot <- nchar(base)
    guard <- 0L
    while (nchar(cur) > width && guard < 500L) {
      guard <- guard + 1L
      window <- substr(cur, 1L, width)
      pos <- which(strsplit(window, "", fixed = TRUE)[[1]] == " ")
      pos <- pos[pos > prot + 1L]
      if (length(pos)) {
        cut <- max(pos)
        out <- c(out, substr(cur, 1L, cut - 1L))
        rest <- substring(cur, cut + 1L)
      } else {
        out <- c(out, window)
        rest <- substring(cur, width + 1L)
      }
      cur <- paste0(cind, rest)
      prot <- nchar(cind)
    }
    out <- c(out, cur)
  }
  out
}

```

A.6 Document chunks

```

knitr::opts_chunk$set(
  echo      = FALSE,
  message   = FALSE,
  warning    = FALSE,
  fig.align = "center",
  fig.pos   = "H",
  out.width = "100%",
  dev       = "pdf",
  cache.path = "cache/",
  fig.path  = "figures/"
)

suppressPackageStartupMessages({
  library(ggplot2)
  library(dplyr)
  library(tidyr)
  library(knitr)
  library(kableExtra)
  library(matrixStats)
  library(quantmod)
  library(tseries)
})

```

```

library(patchwork)
library(scales)
})

source("R/ruin.R")
source("R/checks.R")
source("R/data.R")
source("R/tests.R")
source("R/theme.R")
source("R/utlis.R")

set_gr_defaults(9)
options(knitr.kable.NA = "")

## Parameters for the worked examples quoted in the text. Every number the
## prose reports about these examples is computed from these objects, so the
## text cannot fall out of step with the arithmetic.
ex_N    <- 100L; ex_i  <- ex_N %/% 2L      # fair-game duration example
fm_N    <- 40L; fm_i  <- fm_N %/% 2L      # fundamental matrix example
adv_p    <- 0.49; adv_i <- 100L           # adverse-drift duration example
adv_dur  <- adv_i / ((1 - adv_p) - adv_p)  # limit of  $D_i$  as  $N \rightarrow$  infinity

## Empirical experiment parameters, defined here because the text refers to
## them before the simulation chunk that consumes them.
i_emp    <- 20L      # starting capital, in units of the mean absolute daily move
N_emp    <- 100L     # upper barrier, same units
blockL   <- 10L     # moving-block bootstrap block length, in trading days

## Parameters for the figures, tables and simulations. knitr evaluates a
## caption before it runs the chunk body, so anything a caption quotes has to
## be defined here rather than beside the code that uses it. Keeping them in
## one place is also what stops a caption from disagreeing with the
## computation it describes.
tm_N      <- 6L;    tm_p <- 0.6           # illustrative transition matrix
Nmc       <- 100L   # target capital, numerical sections
curve_ps  <- c(0.44, 0.48, 0.50, 0.52, 0.56) # win rates drawn in Figure 1
curve_hi  <- 0.48   # the win rate singled out in text
hm_level  <- 0.01   # contour level on the heatmap
path_N    <- 40L;   path_i <- 20L; n_paths <- 10L
n_rep     <- 100000L # replications per Monte Carlo cell
conv_i    <- 40L;   conv_n <- 200000L     # convergence study
dur_n     <- 20000L # replications per duration cell
boot_n    <- 50000L # replications per bootstrap arm
sens_n    <- 20000L # replications per block-length row
curve_n   <- 12000L # replications per point in Figure 8
vr_qs     <- c(2L, 4L, 8L, 16L, 32L)      # variance ratio horizons
lb_lags   <- 10L   # lags in the Ljung-Box tests
conv_true <- ruin_prob(conv_i, Nmc, 0.5)

## Long source lines in the appendix listing are wrapped rather than allowed
## to run into the margin. Only chunks marked `wrapsrc = TRUE` are affected.
default_source_hook <- knitr::knit_hooks$get("source")
knitr::knit_hooks$set(source = function(x, options) {
  if (isTRUE(options$wrapsrc)) {
    lines <- unlist(strsplit(paste(x, collapse = "\n"), "\n", fixed = TRUE))
    x <- paste(wrap_src_lines(lines, width = 80L), collapse = "\n")
  }
  default_source_hook(x, options)
})

default_output_hook <- knitr::knit_hooks$get("output")
knitr::knit_hooks$set(output = function(x, options) {
  if (isTRUE(options$wrapout)) {
    lines <- unlist(strsplit(paste(x, collapse = "\n"), "\n", fixed = TRUE))
    x <- paste(wrap_src_lines(lines, width = 88L, cont = "  "), collapse = "\n")
  }
  default_output_hook(x, options)
})

```

```

})
Pm <- transition_matrix(tm_N, tm_p)
dimnames(Pm) <- list(paste0("$j=", 0:tm_N, "$"), paste0("$k=", 0:tm_N, "$"))
kbl(Pm, booktabs = TRUE, digits = 1, escape = FALSE, linesep = "",
    caption = sprintf("Transition matrix for $N = %d$ and $p = %.1f$. The first
        and last rows are absorbing; every interior row places mass $p$ one
        step up and $q$ one step down. Zeros are shown to make the tridiagonal
        band explicit.", tm_N, tm_p)) |>
    kable_styling(latex_options = c("HOLD_position"), font_size = 9)
i0 <- 30; NO <- 100
eps <- 10^-(1:7)
lim_tab <- data.frame(
    eps = eps,
    exact = sapply(eps, function(e) ruin_prob(i0, NO, 0.5 + e)),
    base = i0 / NO
)
lim_tab$err0 <- lim_tab$exact - lim_tab$base
lim_tab$first <- lim_tab$base + 2 * eps * i0 * (NO - i0) / NO
lim_tab$err1 <- lim_tab$exact - lim_tab$first

kbl(data.frame(
    `\\varepsilon` = sprintf("$10^{-%d}$", 1:7),
    `Exact $P_i$` = fmt(lim_tab$exact, 9),
    `$P_i - i/N$` = sprintf("%.3e", lim_tab$err0),
    `Ratio to $2\\varepsilon i(N-i)/N$` = fmt(lim_tab$err0 / (2 * eps * i0 *
        (NO - i0) / NO), 5),
    `Residual after correction` = sprintf("%.3e", lim_tab$err1),
    check.names = FALSE,
    booktabs = TRUE, escape = FALSE, linesep = "", align = "lrrrr",
    caption = sprintf("Convergence of the asymmetric ruin formula to the
        symmetric limit for $i = %d$, $N = %d$. The third column shows the
        deviation from $i/N$ shrinking linearly in $\\varepsilon$; the fourth
        shows that deviation divided by the first-order prediction
        $2\\varepsilon i(N-i)/N = %s$, which tends to one; the fifth shows
        the residual after the first-order term is removed, which shrinks
        quadratically.", i0, NO, fmt(2 * i0 * (NO - i0) / NO))) |>
    kable_styling(latex_options = c("HOLD_position"), font_size = 9)
hy <- huygens_odds()
mart_check <- function(i, N, p) {
    r <- (1 - p) / p
    PN <- ruin_prob(i, N, p) # probability of absorption at N
    Pzero <- ruin_prob_0(i, N, p) # probability of absorption at 0, computed
    # directly rather than as 1 - PN
    lhs <- Pzero * r^0 + PN * r^N # E[r^{S_tau}] implied by the closed form
    rhs <- r^i # what optional stopping requires
    abs(lhs - rhs) / abs(rhs)
}
mart_grid <- expand_grid(i = c(5, 20, 45), N = c(50, 100),
    p = c(0.42, 0.47, 0.53, 0.61))
mart_grid$rel_error <- mapply(mart_check, mart_grid$i,
    mart_grid$N, mart_grid$p)
max_mart_err <- max(mart_grid$rel_error)

chk_martingale <- gr_check(
    "martingale",
    "Optional stopping identity agrees with the difference-equation solution",
    max_mart_err, 1e-12)
cross_cases <- expand_grid(i = c(10, 25, 60), N = c(80, 120),
    p = c(0.44, 0.50, 0.57)) |>
    dplyr::filter(i < N)

cross_tab <- cross_cases |>
    dplyr::rowwise() |>
    dplyr::mutate(
        P_closed = ruin_prob(i, N, p),
        D_closed = ruin_duration(i, N, p),
        P_matrix = matrix_absorption(N, p)$prob_N[as.character(i)],

```

```

    D_matrix = matrix_absorption(N, p)$duration[as.character(i)],
    dP = abs(P_closed - P_matrix),
    dD = abs(D_closed - D_matrix)
  ) |>
  dplyr::ungroup()

max_dP <- max(cross_tab$dP)
# relative error for durations, which span several orders of magnitude
max_dD <- max(cross_tab$dD / pmax(cross_tab$D_closed, 1))

chk_matrix <- gr_check(
  "matrix",
  "Fundamental matrix reproduces both closed forms",
  max(max_dP, max_dD), 1e-10)
kbl(cross_tab |>
  dplyr::transmute(
    `i` = i, `N` = N, `p` = fmt(p, 2),
    `P_i` closed form` = fmt(P_closed, 8),
    `P_i` matrix` = fmt(P_matrix, 8),
    `abs. diff.` = sprintf("%.1e", dP),
    `D_i` closed form` = fmt(D_closed, 3),
    `D_i` matrix` = fmt(D_matrix, 3),
    `rel. diff.` = sprintf("%.1e", dD / pmax(D_closed, 1))),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "rrrrrrrr",
  caption = "Closed-form results compared with the fundamental matrix
  computation. The two routes share no code path: the left-hand values
  come from the algebraic solutions of the difference equations, the
  right-hand values from numerically inverting  $I-Q$ . Agreement is at the
  level of floating-point rounding throughout.") |>
  kable_styling(latex_options = c("HOLD_position", "scale_down"), font_size = 8)

Nc <- Nmc
curve_df <- expand_grid(i = 0:Nc, p = curve_ps) |>
  dplyr::rowwise() |>
  dplyr::mutate(P = ruin_prob(i, Nc, p)) |>
  dplyr::ungroup() |>
  dplyr::mutate(p_lab = factor(sprintf("%.2f", p)))

ggplot(curve_df, aes(i, P, colour = p_lab)) +
  geom_line(linewidth = 0.6) +
  scale_colour_gr(name = "Win probability p") +
  scale_x_continuous(breaks = seq(0, Nc, 20), expand = expansion(mult = 0.01)) +
  scale_y_continuous(labels = label_percent(accuracy = 1),
    breaks = seq(0, 1, 0.25), expand = expansion(mult = 0.02))
  +
  labs(x = "Starting capital i (units)",
    y = "P(reach target before ruin)") +
  theme(legend.position = "right")
deg <- expand_grid(N = c(20, 50, 100, 200, 500, 1000), p = c(0.485, 0.49,
  0.495, 0.50)) |>
  dplyr::rowwise() |>
  dplyr::mutate(i = N / 4, ruin = ruin_prob_0(i, N, p)) |>
  dplyr::ungroup() |>
  dplyr::select(N, p, ruin) |>
  tidyr::pivot_wider(names_from = p, values_from = ruin,
    names_prefix = "p = ")

kbl(deg |> dplyr::mutate(dplyr::across(-N, ~ pct_strict(.x, 2))),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "rrrrr",
  col.names = c("$N$", "$p=0.485$", "$p=0.490$", "$p=0.495$", "$p=0.500$"),
  caption = "Modelled probability of ruin for a gambler starting with a
  quarter of the target, as the target grows. Reading along any row shows
  the cost of a small edge; reading down a column shows that the cost
  compounds with the size of the game. The final column is the fair game,
  where the ruin probability is the constant three quarters.") |>
  kable_styling(latex_options = c("HOLD_position"), font_size = 9)
hm <- expand_grid(p = seq(0.40, 0.60, by = 0.002), i = seq(1, 99, by = 1)) |>
  dplyr::rowwise() |>

```

```

dplyr::mutate(ruin = ruin_prob_0(i, Nmc, p)) |>
dplyr::ungroup()

ggplot(hm, aes(p, i, fill = ruin)) +
  geom_raster(interpolate = TRUE) +
  geom_contour(aes(z = ruin), breaks = hm_level, colour = "white", linewidth =
    0.4) +
  geom_contour(aes(z = ruin), breaks = 0.5, colour = "grey80",
    linewidth = 0.3, linetype = "22") +
  scale_fill_viridis_c(name = "P(ruin)", option = "viridis", direction = -1,
    labels = label_percent(accuracy = 1)) +
  scale_x_continuous(expand = c(0, 0)) +
  scale_y_continuous(expand = c(0, 0)) +
  labs(x = "Win probability p", y = "Starting capital i (units)") +
  theme(legend.position = "right", panel.grid = element_blank())
mu <- 0.30; sigma <- 1.0; x0 <- 3.0; L0 <- 10.0

bm_hit <- function(x, L, mu, sigma) {
  if (mu == 0) x / L else
    (1 - exp(-2 * mu * x / sigma^2)) / (1 - exp(-2 * mu * L / sigma^2))
}

hs <- c(0.5, 0.2, 0.1, 0.05, 0.02, 0.01)
diff_tab <- data.frame(h = hs) |>
dplyr::mutate(
  p_h = 0.5 * (1 + mu * h / sigma^2),
  i = round(x0 / h),
  N = round(L0 / h),
  discrete = mapply(function(i, N, p) ruin_prob(i, N, p), i, N, p_h),
  continuous = bm_hit(x0, L0, mu, sigma),
  abs_error = abs(discrete - continuous)
)

chk_diffusion <- gr_check(
  "diffusion",
  "Discrete ruin probability converges to the Brownian hitting probability",
  diff_tab$abs_error[nrow(diff_tab)], 1e-3,
  pass = diff_tab$abs_error[nrow(diff_tab)] < 1e-3 &&
    all(diff(diff_tab$abs_error) < 1e-9))
kbl(diff_tab |>
  dplyr::transmute(`$h$` = fmt(h, 3), `$p_h$` = fmt(p_h, 5),
    `$i = x/h$` = fmti(i), `$N = L/h$` = fmti(N),
    `Discrete $P_i$` = fmt(discrete, 8),
    `Diffusion limit` = fmt(continuous, 8),
    `Absolute error` = sprintf("%.2e", abs_error)),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "rrrrrrr",
  caption = sprintf("Convergence of the discrete ruin probability to the
    drifted Brownian motion hitting probability, for $\\\\mu = %.2f$,
    $\\\\sigma = %.1f$, $x = %.0f$ and $L = %.0f$. As the stake $h$ shrinks
    the discrete answer approaches the continuous one, with error falling
    roughly in proportion to $h$.", mu, sigma, x0, L0)) |>
  kable_styling(latex_options = c("HOLD_position"), font_size = 9)
set.seed(20260919)

make_paths <- function(p, n_paths, i, N) {
  purrr::map_dfr(seq_len(n_paths), function(k) {
    w <- simulate_walk(i, N, p)
    data.frame(rep = k, step = 0:(length(w$path) - 1), capital = w$path,
      outcome = if (isTRUE(w$hit_N)) "Reached target" else "Ruined")
  })
}

paths_df <- dplyr::bind_rows(
  make_paths(0.50, n_paths, path_i, path_N) |>
  dplyr::mutate(panel = "Fair game, p = 0.50"),
  make_paths(0.45, n_paths, path_i, path_N) |>
  dplyr::mutate(panel = "House edge, p = 0.45")
)

```

```

)
paths_summary <- paths_df |>
  dplyr::group_by(panel, rep) |>
  dplyr::summarise(dur = max(step), outcome = dplyr::first(outcome), .groups =
    "drop")
ggplot(paths_df, aes(step, capital, group = rep, colour = outcome)) +
  geom_hline(yintercept = c(0, path_N), linewidth = 0.45, colour = "grey25") +
  geom_line(linewidth = 0.35, alpha = 0.9) +
  facet_wrap(~ panel, nrow = 1, scales = "free_x") +
  scale_colour_manual(name = NULL, values = c("Reached target" = col_target,
    "Ruined" = col_ruin)) +
  scale_y_continuous(breaks = seq(0, path_N, 10), limits = c(-1, path_N + 1)) +
  labs(x = "Round of play", y = "Capital (units)")
set.seed(20260920)
mc_cells <- expand.grid(i = c(25L, 50L, 75L), p = c(0.48, 0.50, 0.52))

mc_raw <- purrr::pmap(mc_cells, function(i, p) {
  sim <- simulate_ruin(n_rep, i = i, N = Nmc, p = p)
  list(i = i, p = p, hits = sum(sim$hit_upper), mean_dur = mean(sim$duration),
    durations = if (i == 50L && p == 0.50) sim$duration else NULL)
})

focal_durations <- Filter(Negate(is.null), lapply(mc_raw, `[`,
  "durations"))[[1]]

mc_tab <- purrr::map_dfr(mc_raw, function(z) {
  w <- wilson_ci(z$hits, n_rep)
  cp <- clopper_pearson_ci(z$hits, n_rep)
  data.frame(i = z$i, p = z$p, hits = z$hits,
    est = z$hits / n_rep,
    wilson_lo = w$lower, wilson_hi = w$upper,
    cp_lo = cp$lower, cp_hi = cp$upper,
    mean_dur = z$mean_dur)
}) |>
  dplyr::rowwise() |>
  dplyr::mutate(
    exact = ruin_prob(i, Nmc, p),
    exact_dur = ruin_duration(i, Nmc, p),
    abs_err = abs(est - exact),
    se = sqrt(exact * (1 - exact) / n_rep),
    z_score = (est - exact) / se,
    covered = exact >= wilson_lo & exact <= wilson_hi
  ) |>
  dplyr::ungroup()

## The exact value must lie inside a 99.9 per cent Clopper-Pearson interval in
## every cell, the standardised deviations must be of ordinary size, and the
## simulated mean durations must track the closed form.
cp999 <- clopper_pearson_ci(mc_tab$hits, n_rep, conf = 0.999)
chk_mc_cover <- gr_check(
  "mc-coverage",
  "Exact ruin probability inside the 99.9 per cent interval in every cell",
  sum(!(mc_tab$exact >= cp999$lower & mc_tab$exact <= cp999$upper)), 1)
chk_mc_z <- gr_check(
  "mc-z",
  "Largest standardised Monte Carlo deviation within sampling range",
  max(abs(mc_tab$z_score)), 4)
chk_mc_dur <- gr_check(
  "mc-duration",
  "Simulated mean durations within 5 per cent of the closed form",
  max(abs(mc_tab$mean_dur - mc_tab$exact_dur) / pmax(mc_tab$exact_dur, 1)),
  0.05)
kbl(mc_tab |>
  dplyr::transmute(
    `i` = i, `p` = fmt(p, 2),
    `Exact P_i` = fmt(exact, 6),
    `Monte Carlo` = fmt(est, 6),

```

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`Abs. error`      = sprintf("%.2e", abs_err),
`Wilson 95\\% interval` = fmt_ci(wilson_lo, wilson_hi, 6),
`Clopper-Pearson 95\\%` = fmt_ci(cp_lo, cp_hi, 6),
`z$` = fmt(z_score, 2)),
booktabs = TRUE, escape = FALSE, linesep = "", align = "rrrrrccr",
caption = sprintf("Monte Carlo estimates of the probability of reaching the
  target against the closed form, with %s replications per cell and a
  target of $N = %d$ units. The standardised deviation $z$ in the final
  column is the error in units of its own standard error; all nine values
  are within the range expected from sampling variation alone, and every
  interval covers the exact value.", fmti(n_rep), Nmc)) |>
kable_styling(latex_options = c("HOLD_position", "scale_down"), font_size = 8)
set.seed(20260921)
conv_sim <- simulate_ruin(conv_n, i = conv_i, N = Nmc, p = 0.50)

conv_df <- data.frame(n = seq_len(conv_n),
  run = cumsum(conv_sim$hit_upper) / seq_len(conv_n)) |>
dplyr::filter(n >= 100, n %% 25 == 0) |>
dplyr::mutate(se = sqrt(conv_true * (1 - conv_true) / n),
  lo = conv_true - 1.96 * se, hi = conv_true + 1.96 * se)

final_err <- abs(conv_df$run[nrow(conv_df)] - conv_true)
inside <- mean(abs(conv_df$run - conv_true) <= 1.96 * conv_df$se)
chk_convergence <- gr_check(
  "convergence",
  "Running estimate settles on the exact value",
  final_err, 0.01)
ggplot(conv_df, aes(n)) +
  geom_ribbon(aes(ymin = lo, ymax = hi), fill = col_target, alpha = 0.16) +
  geom_hline(yintercept = conv_true, colour = "grey25", linewidth = 0.4) +
  geom_line(aes(y = run), colour = col_ruin, linewidth = 0.35) +
  scale_x_log10(labels = label_comma(),
    breaks = c(100, 1e3, 1e4, 1e5, 2e5)) +
  scale_y_continuous(labels = label_number(accuracy = 0.01)) +
  coord_cartesian(ylim = conv_true + c(-0.06, 0.06)) +
  labs(x = "Number of replications n (log scale)",
    y = "Estimated P(reach target)")
set.seed(20260922)
dur_cells <- expand_grid(i = seq(5L, 95L, by = 10L), p = c(0.48, 0.50))

dur_tab <- purrr::pmap_dfr(dur_cells, function(i, p) {
  sim <- simulate_ruin(dur_n, i = i, N = Nmc, p = p)
  data.frame(i = i, p = p,
    sim_dur = mean(sim$duration),
    sim_se = sd(sim$duration) / sqrt(dur_n),
    exact = ruin_duration(i, Nmc, p))
}) |>
dplyr::mutate(z = (sim_dur - exact) / sim_se)

chk_dur_grid <- gr_check(
  "duration-grid",
  "Simulated durations match the closed form across starting capitals",
  max(abs(dur_tab$z)), 4)
dur_curve <- expand_grid(i = seq(0, Nmc, 0.5), p = c(0.48, 0.50)) |>
dplyr::rowwise() |>
dplyr::mutate(exact = ruin_duration(i, Nmc, p)) |>
dplyr::ungroup()
lab_map <- c("0.48" = "p = 0.48", "0.5" = "p = 0.50")
dur_curve$lab <- lab_map[as.character(dur_curve$p)]
dur_tab$lab <- lab_map[as.character(dur_tab$p)]

ggplot(dur_curve, aes(i, exact, colour = lab)) +
  geom_line(linewidth = 0.6) +
  geom_errorbar(data = dur_tab,
    aes(x = i, ymin = sim_dur - 2 * sim_se, ymax = sim_dur + 2 *
      sim_se),
    inherit.aes = FALSE, width = 1.6, colour = "grey30", linewidth

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```

      = 0.3) +
geom_point(data = dur_tab, aes(i, sim_dur, fill = lab), shape = 21,
           colour = "white", size = 1.9, stroke = 0.3) +
scale_colour_gr(name = NULL) + scale_fill_gr(name = NULL) +
scale_x_continuous(breaks = seq(0, 100, 20)) +
scale_y_continuous(labels = label_comma()) +
labs(x = "Starting capital i (units)", y = "Expected duration (rounds)")
dur_stats <- list(mean = mean(focal_durations), median =
  median(focal_durations),
  q90 = quantile(focal_durations, 0.90),
  q99 = quantile(focal_durations, 0.99), max =
    max(focal_durations))

ggplot(data.frame(d = focal_durations), aes(d)) +
  geom_histogram(bins = 90, fill = col_target, alpha = 0.75, colour = NA) +
  geom_vline(xintercept = dur_stats$mean, colour = col_ruin, linewidth = 0.5) +
  geom_vline(xintercept = dur_stats$median, colour = "grey25",
    linewidth = 0.45, linetype = "22") +
  annotate("text", x = dur_stats$mean, y = Inf, label = " mean",
    hjust = 0, vjust = 1.6, size = 2.7, colour = col_ruin) +
  annotate("text", x = dur_stats$median, y = Inf, label = "median ",
    hjust = 1, vjust = 1.6, size = 2.7, colour = "grey25") +
  scale_x_continuous(labels = label_comma(), limits = c(0,
    quantile(focal_durations, 0.995))) +
  scale_y_continuous(labels = label_comma()) +
  labs(x = "Duration of play (rounds)", y = "Replications")
sp_obj <- load_sp500()
sp <- sp_obj$data
sp$ret <- c(NA, diff(log(sp$close)))
rets <- sp$ret[-1]
T_obs <- length(rets)
n_zero <- sum(rets == 0)
rnz <- rets[rets != 0]
emp_sum <- data.frame(
  Statistic = c("Observations", "Mean daily log return", "Standard deviation",
    "Mean absolute return", "Skewness", "Excess kurtosis",
    "Minimum", "Maximum"),
  Value = c(fmt(T_obs),
    sprintf("%.6f", mean(rets)),
    sprintf("%.6f", sd(rets)),
    sprintf("%.6f", mean(abs(rets))),
    fmt(mean((rets - mean(rets))^3) / sd(rets)^3, 3),
    fmt(mean((rets - mean(rets))^4) / sd(rets)^4 - 3, 3),
    sprintf("%.4f", min(rets)),
    sprintf("%.4f", max(rets)))
)
kbl(emp_sum, booktabs = TRUE, escape = FALSE, linesep = "", align = "lr",
  caption = "Summary statistics for daily S&P 500 log returns over the
  sample period. The excess kurtosis is the quantity of interest for what
  follows: a normal distribution would give zero, and the observed value
  indicates a distribution in which extreme days are far more common than
  any model with fixed step size can accommodate.") |>
  kable_styling(latex_options = c("HOLD_position"), font_size = 9)
n_up <- sum(rnz > 0)
n_nz <- length(rnz)
phat <- n_up / n_nz
w_ci <- wilson_ci(n_up, n_nz)
cp_ci <- clopper_pearson_ci(n_up, n_nz)
signs <- sign(rnz)

rt_own <- runs_summary(signs)
rt_pkg <- tseries::runs.test(factor(signs))
lb_sign <- Box.test(signs, lag = lb_lags, type = "Ljung-Box")
lb_ret <- Box.test(rets, lag = lb_lags, type = "Ljung-Box")
lb_abs <- Box.test(abs(rets), lag = lb_lags, type = "Ljung-Box")
vr_tab <- do.call(rbind, lapply(vr_qs, function(q) lo_mackinlay_vr(rets, q)))
acf1 <- as.numeric(acf(signs, lag.max = 1, plot = FALSE)$acf[2])

```

```

## The independent implementation must agree with the package routine.
chk_runs <- gr_check(
  "runs-test",
  "Own runs-test implementation agrees with tseries::runs.test",
  abs(rt_own$z - as.numeric(rt_pkg$statistic)), 1e-8)
kbl(vr_tab |>
  dplyr::transmute(`$q$` = q,
    `$$\mathrm{VR}(q)$` = fmt(VR, 4),
    `$z_1$ (homoskedastic)` = fmt(z1, 3),
    `$p$-value` = fmt(p1),
    `$z_2$ (robust)` = fmt(z2, 3),
    `$p$-value` = fmt(p2)),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "rrrrrr",
  caption = sprintf("Lo-MacKinlay variance ratio tests on daily log returns
    at aggregation horizons of %d to %d days. A ratio below one indicates
    mean reversion. The homoskedastic statistic $z_1$ and the
    heteroskedasticity-consistent statistic $z_2$ tell noticeably different
    stories, and the robust column is the one to believe for return data.",
    min(vr_qs), max(vr_qs))) |>
  kable_styling(latex_options = c("HOLD_position"), font_size = 9)
set.seed(20260924)
n_cal <- 400L; T_cal <- 2000L; q_cal <- 4L

sim_garch <- function(n, omega = 0.05, alpha = 0.10, beta = 0.85) {
  e <- rnorm(n); s2 <- numeric(n); x <- numeric(n)
  s2[1] <- omega / (1 - alpha - beta); x[1] <- sqrt(s2[1]) * e[1]
  for (t in 2:n) {
    s2[t] <- omega + alpha * x[t - 1]^2 + beta * s2[t - 1]
    x[t] <- sqrt(s2[t]) * e[t]
  }
  x
}

cal_designs <- list(
  `Independent normal` = function() rnorm(T_cal),
  `Independent Student t(3)` = function() rt(T_cal, df = 3),
  `GARCH(1,1), no predictability` = function() sim_garch(T_cal)
)

vr_cal <- purrr::imap_dfr(cal_designs, function(gen, nm) {
  z <- t(replicate(n_cal, {
    u <- lo_mackinlay_vr(gen(), q_cal); c(u$VR, u$z1, u$z2)
  }))
  data.frame(design = nm, mean_vr = mean(z[, 1]),
    rej1 = mean(abs(z[, 2]) > 1.96), rej2 = mean(abs(z[, 3]) > 1.96),
    sd2 = sd(z[, 3]))
})

## Under independence the robust statistic must hold its nominal size, and
## under clustering it must hold it where the homoskedastic one does not.
chk_vr_size <- gr_check(
  "vr-size",
  "Robust variance ratio statistic holds nominal size under independence",
  abs(vr_cal$rej2[1] - 0.05), 0.03)
chk_vr_robust <- gr_check(
  "vr-robust",
  "Robust rejection rate does not exceed the homoskedastic one under
    clustering",
  vr_cal$rej2[3] - vr_cal$rej1[3], 0.01)
kbl(vr_cal |>
  dplyr::transmute(`Data generating process` = design,
    `Mean $$\mathrm{VR}(4)$` = fmt(mean_vr, 4),
    `Rejection rate, $z_1$` = pct(rej1, 1),
    `Rejection rate, $z_2$` = pct(rej2, 1)),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "lrrr",
  caption = sprintf("Calibration of the variance ratio implementation on %s
    simulated samples of %s observations each, none of which contains any

```

```

    predictability. A correctly sized test rejects 5 per cent of the time.
    Under volatility clustering the homoskedastic statistic rejects far too
    often while the robust statistic holds its size, which is the behaviour
    the robust correction exists to deliver and is the evidence that the
    implementation is correct.", fmti(n_cal), fmti(T_cal))) |>
kable_styling(latex_options = c("HOLD_position"), font_size = 9)
test_sum <- data.frame(
  Test = c("Proportion of up days ( $\hat{p} = 0.5$ ?)",
    "Wald-Wolfowitz runs test on signs",
    sprintf("Ljung-Box (%d lags) on signs", lb_lags),
    sprintf("Ljung-Box (%d lags) on returns", lb_lags),
    sprintf("Ljung-Box (%d lags) on absolute returns", lb_lags),
    sprintf("Variance ratio,  $q=d$  (robust)", min(vr_qs)),
    sprintf("Variance ratio,  $q=d$  (robust)", max(vr_qs))),
  Statistic = c(fmt((phat - 0.5) / sqrt(0.25 / n_nz), 3),
    fmt(rt_own$z, 3),
    fmt(as.numeric(lb_sign$statistic), 2),
    fmt(as.numeric(lb_ret$statistic), 2),
    fmti(as.numeric(lb_abs$statistic)),
    fmt(vr_tab$z2[1], 3),
    fmt(vr_tab$z2[length(vr_qs)], 3)),
  `p-value` = c(fmt(p2 * pnorm(-abs((phat - 0.5) / sqrt(0.25 / n_nz))),
    fmt(p2_own$p_value),
    fmt(p2_sign$p_value),
    fmt(p2_ret$p_value),
    fmt(p2_abs$p_value),
    fmt(vr_tab$p2[1]),
    fmt(vr_tab$p2[length(vr_qs)])),
  Conclusion = c("Reject fairness", "Reject independence", "Reject
    independence",
    "Reject independence", "Strong dependence in magnitudes",
    "Reject random walk", "Does not reject at 5%"),
  check.names = FALSE
)
kbl(test_sum, booktabs = TRUE, escape = FALSE, linesep = "", align = "lrrl",
  caption = "Summary of tests of the random walk hypothesis. The statistical
    verdict is a rejection, but the magnitudes involved are small, and the
    strongest dependence in the data is in the size of returns rather than
    in their direction.") |>
kable_styling(latex_options = c("HOLD_position", "scale_down"), font_size = 8)
unit_abs <- mean(abs(rets))
std_step <- rets / unit_abs

acf_r <- acf(rets, lag.max = 60, plot = FALSE)
acf_a <- acf(abs(rets), lag.max = 60, plot = FALSE)
acf_df <- rbind(
  data.frame(lag = as.numeric(acf_r$lag), acf = as.numeric(acf_r$acf), which =
    "Returns"),
  data.frame(lag = as.numeric(acf_a$lag), acf = as.numeric(acf_a$acf), which =
    "Absolute returns")
) |> dplyr::filter(lag > 0)
ci_band <- 1.96 / sqrt(T_obs)

pA <- ggplot(sp[-1, ], aes(date, ret)) +
  geom_line(linewidth = 0.16, colour = col_neutral) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  labs(x = NULL, y = "Daily log return", title = "A. Return series")

pB <- ggplot(acf_df, aes(lag, acf, colour = which)) +
  geom_hline(yintercept = c(-ci_band, ci_band), linetype = "22",
    colour = "grey55", linewidth = 0.3) +
  geom_hline(yintercept = 0, colour = "grey30", linewidth = 0.3) +
  geom_line(linewidth = 0.45) +
  scale_colour_manual(name = NULL, values = c("Returns" = col_target,
    "Absolute returns" = col_ruin)) +
  labs(x = "Lag (trading days)", y = "Autocorrelation",
    title = "B. Autocorrelation structure")

```

```

qq <- data.frame(theo = qnorm(ppoints(T_obs)) * sd(rets) + mean(rets),
  emp = sort(rets))
pC <- ggplot(qq, aes(theo, emp)) +
  geom_abline(slope = 1, intercept = 0, colour = col_ruin, linewidth = 0.4) +
  geom_point(size = 0.25, colour = col_target, alpha = 0.5) +
  scale_x_continuous(labels = label_percent(accuracy = 1)) +
  scale_y_continuous(labels = label_percent(accuracy = 1)) +
  labs(x = "Normal quantile", y = "Empirical quantile",
    title = "C. Normal quantile-quantile plot")

pD <- ggplot(data.frame(s = std_step[abs(std_step) < 6]), aes(s)) +
  geom_histogram(bins = 90, fill = col_target, alpha = 0.7, colour = NA) +
  geom_vline(xintercept = c(-1, 1), colour = col_ruin, linewidth = 0.5) +
  annotate("text", x = 1, y = Inf, label = " model step ", hjust = 0, vjust =
    1.8,
    size = 2.6, colour = col_ruin) +
  labs(x = "Daily move in units of the mean absolute move",
    y = "Days", title = "D. Actual step sizes against the model's")

(pA / pB) | (pC / pD)
cap_p <- c(0.5100, 0.5200, 0.5300, round(phat, 4), 0.5450, 0.5600)
cap_tab <- data.frame(p = sort(unique(cap_p))) |>
  dplyr::rowwise() |>
  dplyr::mutate(
    edge = 2 * p - 1,
    i_1pct = capital_for_ruin_target(p, 0.01, N = Inf),
    i_5pct = capital_for_ruin_target(p, 0.05, N = Inf),
    exact_1 = log(0.01) / log((1 - p) / p),
    ruin_at_20 = ruin_prob_infinite(20, p)
  ) |>
  dplyr::ungroup()

req_cap <- cap_tab$i_1pct[which.min(abs(cap_tab$p - round(phat, 4)))]

kbl(cap_tab |>
  dplyr::transmute(
    `p` = fmt(p, 4),
    `Edge 2p-1` = pct(edge, 2),
    `Units for 5%% ruin` = fmti(i_5pct),
    `Units for 1%% ruin` = fmti(i_1pct),
    `Exact log(alpha)/log(q/p)` = fmt(exact_1, 2),
    `Ruin with 20 units` = pct(ruin_at_20, 2)),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "rrrrrr",
  caption = sprintf("Capital required for the modelled probability of
    eventual ruin to fall below 5 and 1 per cent, against an opponent of
    unlimited capital, for win rates spanning the estimate %\\hat p =
    %s. One unit is the amount staked per day. The final column shows how
    exposed a trader with 20 units would be at each win rate.", fmt(phat,
    4))) |>
  kable_styling(latex_options = c("HOLD_position", "scale_down"), font_size = 8)
set.seed(20260923)

boot_arms <- list(
  `Idealised +/-1 model` = bernoulli_sampler(phat),
  `IID bootstrap` = iid_boot_sampler(std_step),
  `Block bootstrap` = block_boot_sampler(std_step, blockL)
)

boot_res <- purrr::imap_dfr(boot_arms, function(smp, nm) {
  sim <- simulate_absorbed(boot_n, start = i_emp, upper = N_emp,
    step_sampler = smp, max_steps = 2e5)
  ci <- wilson_ci(sum(!sim$hit_upper), boot_n)
  data.frame(arm = nm, ruin = 1 - mean(sim$hit_upper),
    lo = ci$lower, hi = ci$upper,
    mean_dur = mean(sim$duration),
    median_dur = median(sim$duration))
})

```

```

## Sensitivity of the block bootstrap to the block length.
block_sens <- purrr::map_dfr(c(1L, 5L, 10L, 20L, 40L, 60L), function(L) {
  smp <- if (L == 1L) iid_boot_sampler(std_step) else
    block_boot_sampler(std_step, L)
  sim <- simulate_absorbed(sens_n, start = i_emp, upper = N_emp,
    step_sampler = smp, max_steps = 2e5)
  data.frame(L = L, ruin = 1 - mean(sim$hit_upper),
    se = sd(!sim$hit_upper) / sqrt(sens_n))
})

## Ruin probability against starting capital, all three arms.
boot_curve <- purrr::imap_dfr(boot_arms, function(smp, nm) {
  purrr::map_dfr(seq(5L, 60L, by = 5L), function(i0) {
    sim <- simulate_absorbed(curve_n, start = i0, upper = N_emp,
      step_sampler = smp, max_steps = 2e5)
    data.frame(arm = nm, i = i0, ruin = 1 - mean(sim$hit_upper))
  })
})

ideal_closed <- ruin_prob_0(i_emp, N_emp, phat)
chk_bootstrap <- gr_check(
  "bootstrap-baseline",
  "Simulated idealised arm reproduces its own closed form",
  abs(boot_res$ruin[boot_res$arm == "Idealised +/-1 model"] - ideal_closed),
  0.01)
kbl(boot_res |>
  dplyr::transmute(
    Specification = arm,
    `Modelled ruin probability` = pct(ruin, 2),
    `Wilson 95\\% interval` = paste0("[", pct(lo, 2), ", ", pct(hi, 2),
      "]" ),
    `Mean duration (days)` = fmti(mean_dur),
    `Median duration (days)` = fmti(median_dur)),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "lrrrr",
  caption = sprintf("Projected probability of ruin for a trader starting with
    %d units and a target of %d units, where one unit is the mean absolute
    daily move. All three specifications use the same simulation engine,
    the same barriers and the same mean absolute step size; they differ
    only in the distribution of the daily step. The idealised row agrees
    with its closed form to within simulation error.", i_emp, N_emp)) |>
  kable_styling(latex_options = c("HOLD_position", "scale_down"), font_size = 8)
ideal_line <- data.frame(i = seq(1, 60, 0.5)) |>
  dplyr::mutate(ruin = ruin_prob_0(i, N_emp, phat), arm = "Idealised +/-1
    model")

ggplot(boot_curve, aes(i, ruin, colour = arm)) +
  geom_line(data = ideal_line, linewidth = 0.5) +
  geom_point(size = 1.4) +
  geom_line(data = dplyr::filter(boot_curve, arm != "Idealised +/-1 model"),
    linewidth = 0.4, alpha = 0.7) +
  scale_colour_gr(name = NULL) +
  scale_y_continuous(labels = label_percent(accuracy = 1), limits = c(0, 1)) +
  scale_x_continuous(breaks = seq(0, 60, 10)) +
  labs(x = "Starting capital i (units of the mean absolute daily move)",
    y = "Projected probability of ruin")
kbl(block_sens |>
  dplyr::transmute(`Block length $L$ (days)` = ifelse(L == 1, "1 (iid)",
    as.character(L)),
    `Projected ruin probability` = pct(ruin, 2),
    `Standard error` = pct(se, 2)),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "rrr",
  caption = sprintf("Sensitivity of the block bootstrap to the block length,
    at %s replications per row. The projection is stable for blocks of %d
    days or more and consistently below the independent bootstrap, so the
    direction of the effect is not an artefact of the particular block
    length chosen.", fmti(sens_n), min(block_sens$L[block_sens$L > 1])) |>
  kable_styling(latex_options = c("HOLD_position"), font_size = 9)

```

```

reflecting_stationary <- function(N, p) {
  q <- 1 - p
  Pr <- matrix(0, N + 1, N + 1)
  Pr[1, 1] <- q; Pr[1, 2] <- p # reflection at 0
  Pr[N + 1, N + 1] <- p; Pr[N + 1, N] <- q # reflection at N
  for (k in 1:(N - 1)) { Pr[k + 1, k + 2] <- p; Pr[k + 1, k] <- q }
  ev <- eigen(t(Pr))
  pi_ <- Re(ev$vectors[, which.min(abs(ev$values - 1))])
  pi_ / sum(pi_)
}

N_r <- 12; p_r <- 0.55
pi_numeric <- reflecting_stationary(N_r, p_r)
pi_formula <- (p_r / (1 - p_r))^(0:N_r) / sum((p_r / (1 - p_r))^(0:N_r))
chk_reflecting <- gr_check(
  "reflecting",
  "Numerical stationary distribution matches the truncated geometric",
  max(abs(pi_numeric - pi_formula)), 1e-10)
all_checks <- gr_summary(
  chk_martingale, chk_matrix, chk_diffusion,
  chk_mc_cover, chk_mc_z, chk_mc_dur, chk_convergence, chk_dur_grid,
  chk_runs, chk_vr_size, chk_vr_robust, chk_bootstrap, chk_reflecting)

kbl(all_checks |>
  dplyr::transmute(
    Check = id,
    `What it establishes` = description,
    Observed = sprintf("%.2e", observed),
    Tolerance = sprintf("%.0e", tolerance),
    Status = ifelse(status == "pass", "pass", "\\textbf{FAIL}")),
  booktabs = TRUE, escape = FALSE, linesep = "", align = "llrrr",
  caption = sprintf("The %d verification checks performed while compiling
    this report, with the tolerance each was held to and the value actually
    observed. Any failure is listed first, and the quantity depending on it
    is withheld from the text.", nrow(all_checks))) |>
  kable_styling(latex_options = c("HOLD_position", "scale_down"), font_size = 8)

```

B Appendix B: Session information

The following records the exact computing environment in which this document was compiled, so that the results can be reproduced or, failing that, the discrepancy attributed.

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24)
## Platform: aarch64-apple-darwin23
## Running under: macOS Tahoe 26.6.2
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;
## LAPACK version 3.12.1
##
## locale:
## [1] en_CA/en_CA/en_CA/C/en_CA/en_CA
##
## time zone: America/Toronto
## tzcode source: internal
##
## attached base packages:
## [1] grid      stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] scales_1.4.0      patchwork_1.3.2    tseries_0.10-63    quantmod_0.4.29
## [5] TTR_0.24.4        xts_0.14.2         zoo_1.9-0          matrixStats_1.5.0
## [9] kableExtra_1.4.1  knitr_1.51         tidyr_1.3.2        dplyr_1.2.1
## [13] ggplot2_4.0.3
##
## loaded via a namespace (and not attached):
## [1] generics_0.1.4    xml2_1.6.0         stringi_1.8.9      lattice_0.22-9
## [5] digest_0.6.39     magrittr_2.0.5     evaluate_1.0.5     RColorBrewer_1.1-3
## [9] bookdown_0.48     fastmap_1.2.0      purrr_1.2.2        viridisLite_0.4.3
## [13] codetools_0.2-20  isoband_0.3.0      textshaping_1.0.5  cli_3.6.6
## [17] rlang_1.3.0       withr_3.0.3        yaml_2.3.12        tools_4.6.1
## [21] curl_8.0.0        vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5
## [25] stringr_1.6.0     pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6
## [29] glue_1.8.1        systemfonts_1.3.2  xfun_0.60          tibble_3.3.1
## [33] tidysselect_1.2.1 rstudioapi_0.19.0  farver_2.1.2       htmltools_0.5.9
## [37] rmarkdown_2.32    svglite_2.2.2      labeling_0.4.3     compiler_4.6.1
## [41] S7_0.2.2          quadprog_1.5-8
```